Multidimensional Lattice Walk Enumeration Through Coefficient Extraction Operators (DRAFT 1)

by

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Abstract

In this thesis we investigate the application of an operator approach to the enumeration of lattice walks in multiple dimensions, in order to automate some of the analysis of finding the generating function Q for the walks. In particular, given a lattice walk model in d dimensions with an arbitrary step set, some interactions and some regional constraints, we can write down a functional equation $(1 - t\Gamma S)Q = q$, where Γ is an operator involving coefficient extractions. We also use the operator approach to unify and extend the algebraic and obstinate kernel methods. Most of the thesis focuses on models with small steps confined to the an orthant and covers generalizations in the end.

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Preface

The research and the write-up for this thesis is wholly done by me, Aleksandar Vlasev, with some collaboration and mentoring from my advisor Andrew Rechnitzer. In particular, Andrew introduced me to the idea of studying lattice walks with interactions, where he and his colleagues have studied such problems to various success, and provided his invaluable guidance along the way. I did all the work and most of it is new, to the best of my knowledge. Whenever it was not, I have provided full attribution and references, again, to the best of my knowledge.

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Chapter 1

Introduction

The subject of the enumeration of lattice walk models is a classic combinatorial one, but it has seen a flurry of research in recent years, including the full classification of the generating functions for models with small steps on the quarter plane, and partial classification of their three-dimensional equivalent. There are many other formulations, including self-avoiding walks, walks with interactions with the lattice and with themselves, and so on. Furthermore, there are ideas about automatizing the derivations of equations and generating functions. It appears that we are approaching a more general theory that contains all of these disparate topics under one umbrella. In this thesis, we will provide more evidence that such a theory may exist and we will do so by recasting several methods and stages of analysis into a language of operators. Furthermore, the change of perspective allows for the quick formulation of functional equations for a broad class of lattice walk models. This makes it possible to automatically derive explicit functional equations for models that are nice enough.

Apart from being an interesting mathematical construct, lattice walk models enjoy some applications outside of mathematics. There are parts of the physical world that can be modelled via walk models — for example, various proteins and polymers, although in those cases the models can be more complicated. There is a rich theory of continuous and discrete lattice walks. Specifically, in lattice walk enumeration, we count the number of walks with certain characteristics subject to a multitude of constraints. The topic often leads to the use of generating functions as a powerful tool in the analysis. In the end, we gain insight into what makes certain lattice walk models easier and others more difficult. What follows is a brief recollection of the recent history and state of the art of the problem.

1.1 Recent History

One may start with lattice walks that do not have any constraints — the walks may go anywhere, while taking whatever steps are available. The enumeration does not even need generating functions — just construct all possible sequences of steps from a set $\mathcal{S}$ and then you will find that the number of walks of length n is just $|\mathcal{S}|^n$. In more complicated models we make use of the generating function. To help us, we define a *step polynomial* S keeping inventory of the step set $\mathcal{S}$.

The next step up in difficulty is to consider walks on half-spaces, where walks are not allowed to venture past points with one coordinate equal to zero. These cases are not as simple to solve but various methods allow us to solve them. Check [3–6] for work in one or more dimensions. It turns out that the generating functions we obtain are algebraic. Naturally, we are led to consider cases where the region is an intersection of half-spaces. Even for two dimensions, the problem has generated a rich theory with many different cases — sometimes the generating functions are not even D-finite, let alone algebraic. The main object of study in these cases is a group G of bi-rational transformations preserving the polynomial $K = 1 - tS$, called the *kernel* of the model. For a classification of two-dimensional walks with small steps in the quarter plane, see [8].

When G is finite, in the 23 non-equivalent cases, we obtain a D-finite generating function, as can be seen in the classifications [7–10] for some specific cases. The main tool in the study of the finite group cases is the *algebraic kernel* method where we sum different functional equations produced by group actions by elements of G. This is done to cancel the majority of the terms in the functional equation and allow us to find out information about the model directly. We can also look beyond small steps. For example, see [11] for a treatment of models with finite groups and repeated steps.

There are 56 non-equivalent cases of quarter plane walks with an infinite group. For 51 of them, the generating function is not D-finite [13] . The remaining five cases required a fair bit more work [14, 15]. The main tool used to tackle these five cases is the *iterated kernel* method where we use properties of the roots of the kernel K to show the generating function is not D-finite.

After the classification was completed, we were naturally led to investigate higher-dimensional variants of this problem. In three dimensions we have the recent partial classification [18] of lattice walk models with up to six steps, where the authors extend the algebraic kernel method to handle the majority of finite group cases encountered. However, we do not need to confined our analysis on the positive orthant. For example, see [16, 17] for analysis of models where the region of interest is a Weyl chamber.

Another generalization is to consider walks with interactions and/or self-avoiding walks (or SAWs). These two types of generalizations have important applications in physics, because they can be used to model polymers, proteins and other chains. For some research on SAWs with various

interactions, see [19–25]. The topic of SAWs does not lend itself as nicely to using generating functions like the ones considered so far, so we make simplifications. First, we can consider lattice models like before but with some interactions. For example, see [26, 27] for walks with interactions in an octant. See [28–30] for walks with interactions on the plane, where the authors used the *obstinate* kernel method to find relationships between walks with interactions and walks without them.

There has also been progress towards the automation of the classification of random walk models. For example, see [34] for some classification and [35] for analysis of some of the computational challenges involved in automation. There is also automation in the derivation of asymptotics for some models, for example [37, 38]. A lot of the works mentioned so far have also discussed asymptotics.Since the enumeration of lattice walks is a combinatorial subject, there are connections with automation in combinatorics too. For examples, see [31, 32] for details on symmetric functions, and [33] for a holonomic approach to combinatorics.

Since there is such a rich theory of lattice walks, we hope that there is some kind of general theory unifying the various kernel methods and approaches to solving lattice walk models. By recasting some of the analysis in terms of operators, this thesis provides further evidence that such a theory may exist. The optimistic thing would be to suppose that a general algorithm exists that takes an arbitrary lattice walk model, in arbitrarily many dimensions, with arbitrary finite step sets and with arbitrary constraints; and outputs various data about the model: functional equation for the generating function, a solution and information about the model's D-finiteness. It is only logical that, if such an algorithm were to exist, it would not have difficulty with the particulars of the lattice walk model. While this kind of algorithm may be too strong to envision, it may be the case that there are atrocious lattice walk models that we cannot solve, the idea of the existence of such an algorithm has been a strong motivator in our pursuit of generalizations. This thesis succeeds in demonstrating that the first task is achievable. For a vast assortment of lattice walk models, we can almost-automatically write down a functional equation for its generating function counting length, position and interactions. The equation looks like

$$(1 - t\Gamma S)Q = q\,, \tag{1.1}$$

where Γ is an operator on the variables of Q, S is the Laurent step polynomial keeping inventory of the steps in polynomial form, and q is some Laurent polynomial, serving as an initial condition.

1.2 Roadmap

In this thesis we will work towards a few different goals.

1. Begin the classification of walks with small steps on the quarter plane but with interactions.

2. Extend the algebraic and obstinate kernel methods to models with interactions and higher dimensions.

3. Automatically derive functional equations for higher-dimensional models with various restrictions

In Chapter 2, we will study the enumeration of one-dimensional lattice walk models with small steps and interactions with the origin. We will derive explicit generating functions for all models and the analysis will show the first examples of the methods that we will use in later chapters. For example, we will see how operators make the derivations easier.

In Chapter 3, we will us the powerful operator perspective to derive functional equations for the generating function enumerating higher-dimensional lattice walk models with small steps and interactions with the bounding hyperplanes. In the latter part of the chapter, we will focus on the classification of two-dimensional models and some first solutions. This chapter contains most of the definitions relevant in the analysis.

In Chapter 4, we will recast some of the algebraic kernel analysis in terms of operators and extend it to handle lattice walk models with interactions. We will focus on two-dimensional models with finite groups and see that if the method extends, we will have some tighter constraints. In the end, we are able to handle highly-symmetric models with group of order four, half of the models with group of order six, and all models with group of order eight.

In Chapter 5, we will investigate the obstinate kernel method for the two-dimensional models with interactions and finite groups. Again, we will recast the method in operator notation. For the interacting cases, the algebraic and the obstinate kernel methods are very similar. They can both be understood in the language of weighted orbital operators.

In Chapter 6, we will finally investigate mode general random walks models. That is, we will find an operator functional equation for a large class of random walk models — those with general regional and marking constraints, general step sets and general starting conditions. The difficulty lies in finding a good representation for the operators involved.

Chapter 2

One-dimensional Lattice Walk Models

2.1 Derivation of the Functional Equation

Our analysis starts with the enumeration of one-dimensional lattice walk models to illustrate the methods and ideas that we will employ throughout the thesis. As part of the solutions, we will use operators to simplify some of the work. We will focus on models with interactions and small steps, that is, models where the step size is at most one and we count interactions with the origin. The step set $\mathcal{S} \subseteq \{\text{-}1, 0, 1\}$ can be represented with the polynomial

$$S = S_{\text{-}1}\overline{x} + S_0 + S_1 x\,, \tag{2.1}$$

where S_k is 1 or 0 if k is in $\mathcal{S}$ or not, respectively.

We will first look at a basic way of generating walks and then we will find a way to encapsulate the process into an algebraic method. If we start with the empty walk at the origin, we can grow it by taking one of the steps from $\mathcal{S}$ to obtain three walks. One of them ends on the negative axis, so we remove it. We can continue growing the remaining two walks, each time producing a few more walks and removing those that end on negative coordinates. At each step, the length of a walk increases by one and if the walk steps on the origin, we increment a counter tracking the interaction. We can generate all the lattice walks in this way.

We may translate each of the steps in the process into some kind of transformation of polynomials, and this enables us to use powerful mathematical theories to extract information about the properties of lattice walks. We will track the length of walks via powers of t, their interactions via powers of a, and finally, their endpoints via powers of x. The empty walk has length zero and ends at the origin, without interacting. Therefore, its corresponding monomial is $a^0x^0t^0 = 1$. Each of

the steps changes the end coordinate and this corresponds to multiplying the monomial by one of $\{\overline{x}, 1, x\}$. The two surviving walks have monomials at and xt. It is a good exercise to check that the next iteration of this algorithm yields the monomials a^2t^2, axt^2, at^2, xt^2 and x^2t^2. If $\mathcal{S}$ is not the full set, we have fewer walks.

Instead of working with the monomials for individual walks, it would be better to work with the whole set at the same time. To make computations easier, we keep all of the monomials inside a generating function $Q(a, x; t)$ given by

$$Q = Q(x, a; t) = \sum_{k=0}^{\infty} q_k(x, a)t^k, \tag{2.2}$$

where q_k is the generating function for walks of length k. Since extending walks corresponds to multiplication of monomials for walks with monomials for steps from S. Let us see what happens when we multiply q_k by S. When we expand this product, we obtain all combinations of a walk of length k and possible extension via a step from $\mathcal{S}$. That is, we have extended all walks simultaneously. As in the algorithm above, we need a way to eliminate those walks from Sq_k that step on negative coordinates and to multiply by a those walks that step on the origin. The next important idea, and it is the kernel of this thesis, is to use some unknown operator Γ that will help us do just that — remove walks ending on negative coordinates and increment powers of a by one, for those that end on the origin. Then we may simply write $q_{k+1} = \Gamma(Sq_k)$ and sum over powers of t as follows:

$$\sum_{k=0}^{\infty} q_{k+1}t^{k+1} = t\sum_{k=0}^{\infty} \Gamma(Sq_k)t^k = t\Gamma\left(S\sum_{k=0}^{\infty} q_k t^k\right). \tag{2.3}$$

The expression on the left is $Q - q_0 = Q - 1$ and the expression on the right is $t\Gamma(SQ)$. This gives us a functional equation for Q in terms of S and the operator Γ. That is,

$$Q = 1 + t\Gamma SQ\,, \tag{2.4}$$

which we may rewrite as $(1 - t\Gamma S)Q = 1$ to highlight the operator form.

To find the operator Γ, note that when we multiply q_k by S, the farthest a walk can go on the negative axis is the coordinate x = -1, so it is enough to remove terms from Sq_k with power of -1 of x. For the interaction with the origin, we need to multiply the monomial without x by a. All this can be achieved via the coefficient extraction operators $[\overline{x}]$ and $[x^0]$. We may write

$$\Gamma = 1 - \overline{x}[\overline{x}] + (a - 1)[x^0]\,, \tag{2.5}$$

where 1 is the identity operator. When we apply it on some Laurent polynomial F, the identity makes a copy of F, the $-\overline{x}[\overline{x}]$ part removes the $\overline{x}$ term, and the $(a-1)[x^0]$ part multiplies the coefficient in front of the x^0 term by a. This completes our derivation of the functional equation. We will make the extractions explicit later on, but for now, we will look further into the operator Γ.

Suppose that we decided to obtain the operator step-by-step, where we obtain two operators E and L that remove monomials with $\overline{x}$ and mark monomials with x^0, respectively. Then it is reasonable to expect that $\Gamma = LE$ or $\Gamma = EL$ by composition. The operators are given by

$$E = 1 - \overline{x}[\overline{x}] \quad \text{and} \quad L = 1 + (a-1)[x^0]\,, \tag{2.6}$$

and if we let $b = a - 1$ and multiply them out, we find that

$$LE = (1 - \overline{x}[\overline{x}])(1 + b[x^0]) = 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] = 1 + b[x^0] - \overline{x}[\overline{x}]\,. \tag{2.7}$$

Therefore $\Gamma = LE$. Similarly, $\Gamma = EL$, so the operators commute. The last term in the calculation vanished because we use $[x^0]$, the resulting term is constant with respect to x, and the extraction $[\overline{x}]$ yields nothing. Here is how the operators work in a lattice model.

Example 2.1. Suppose that $S = x + \overline{x}$. Starting with $Q_0 = 1$, we have

$$\begin{aligned}
Q_1 = LES(1) &= (1 + b[x^0] - \overline{x}[\overline{x}])(x + \overline{x}) = x + \overline{x} - \overline{x} = x && (2.8)\\
Q_2 = LES(Q_1) &= (1 + b[x^0] - \overline{x}[\overline{x}])(x^2 + 1) = x^2 + 1 + b = a + x^2 && (2.9)\\
Q_3 = LES(Q_2) &= (1 + b[x^0] - \overline{x}[\overline{x}])(a\overline{x} + ax + x + x^3) \\
&= a\overline{x} + ax + x + x^3 - a\overline{x} && (2.10)\\
&= (1+a)x + x^3\,. && (2.11)
\end{aligned}$$

Therefore,

$$\begin{aligned}
Q &= Q_0 + Q_1 t + Q_2 t^2 + Q_3 t^3 + \cdots \\
&= 1 + xt + (a + x^2)t^2 + (ax + x + x^3)t^3 + \cdots\,.
\end{aligned} \tag{2.12}$$

Note that if we apply Γ to tSQ, we obtain Q again but with the first term, $Q_0 = 1$, missing.

The theory we have presented so far is consistent with present methods for finding the functional equation for Q because once we compute the explicit extractions in ΓSQ, we would obtain the classic explicit functional equations. However, the difference lies in the ease of use of operators like Γ versus the painstaking inclusion-exclusion process, which becomes unwieldy for models with interactions in higher dimensions.

To show this consistency, we will perform the explicit extractions. Let $Q_k = [x^k]Q$. Then,

$$\begin{aligned}[x^0]SQ &= [x^0]((S_{\text{-}1}\overline{x} + S_0 + S_1 x)Q) \\ &= (S_{\text{-}1}[x] + S_0[x^0] + S_1[\overline{x}])Q \\ &= S_{\text{-}1}Q_1 + S_0 Q_0\,,\end{aligned} \tag{2.13}$$

where last equality follows since $Q_{\text{-}1} = 0$. Here we used the property that as an operator $[x^i]x^j = [x^{i-j}]$. A similar computation shows that $[\overline{x}]SQ = S_{\text{-}1}Q_0$ and the explicit equation is given by

$$\begin{aligned}1 &= Q - t\Gamma SQ \\ &= Q - t(SQ + b[x^0]SQ - \overline{x}[\overline{x}]S)Q \\ &= Q - t(SQ + b(S_{\text{-}1}Q_1 + S_0Q_0) - \overline{x}S_{\text{-}1}Q_0)\,.\end{aligned} \tag{2.14}$$

We can rewrite it in the following form:

$$(1 - tS)Q - t(bS_0 - \overline{x}S_{\text{-}1})Q_0 - btS_{\text{-}1}Q_1 = 1\,. \tag{2.15}$$

Here $Q = Q(x)$, $Q_0 = Q(0)$ and $Q_1 = Q'(0)$. It turns out that we can eliminate Q_1 by finding an equation for it via applying $[x^0]$ to both sides of the explicit equation. That is,

$$\begin{aligned}1 &= Q_1^0 - t(S_{\text{-}1}Q_1 + S_0Q_0) - tbS_0Q_0 - btS_{\text{-}1}Q_1 \\ &= (1 - atS_0)Q_0 - atS_{\text{-}1}Q_1\,.\end{aligned} \tag{2.16}$$

and substitute this solution for $tS_{\text{-}1}Q_1$ in the previous equation. After some simplification, we obtain the cleaner master functional equation

$$(1 - tS)Q - (c - t\overline{x}S_{\text{-}1})Q_0 = \overline{a}\,. \tag{2.17}$$

We can take a shortcut that avoids this extraction and substitution. The operator L is actually invertible. To undo the multiplication with a, we just need to multiply terms by $1/a$. Therefore, the inverse is $L^{-1} = 1 - c[x^0]$, where $c = 1 - \overline{a}$. This allows us to write the functional equation as

$$(L^{-1} - tES)Q = \overline{a} \tag{2.18}$$

and when we perform the extractions, we obtain Equation 2.17 directly. The left-hand side is

$$L^{-1}Q - tESQ = (Q - cQ_0) - t(SQ - \overline{x}S_{\text{-}1}Q_0) = (1 - tS)Q - (c - t\overline{x}S_{\text{-}1})Q_0\,. \tag{2.19}$$

It might seem like a similar amount of work, but in higher dimensions we would have to solve 2^d equation by back-substitution to obtain the cleaner functional equation. Using L^{-1} lets us avoid all that unnecessary work.For example, in [21, 30] the authors had to perform those back-substitutions for a two-dimensional model. We close this part of the analysis with our first result.

Proposition 2.2. *Given a one-dimensional lattice walk model with small steps and restricted to the non-negative axis, the generating function Q satisfies the functional equations*

$$(L^{-1} - tES)Q = \overline{a}\,, \tag{2.20a}$$

$$(1 - tS)Q - (c - t\overline{x}S_{-1})Q_0 = \overline{a}\,. \tag{2.20b}$$

The first one we call an operator equation, and the second, master functional equation.

The operator notation/perspective is powerful. With it, we can derive functional equations for many lattice walk models in arbitrary dimensions with ease. Usually, derivations have employed inclusion-exclusion and careful book-keeping (Lemma 4 in [8, p. 10], [15, p. 58]) to write master functional equations, or their precursors, in a direct manner. Unfortunately, that approach is difficult to generalize to arbitrary dimensions and more difficult restrictions/marking.

We can also attend to coefficient extraction in a neater way. Instead of extracting $[x^k]$ from the master functional equation, it is better to first go through the operator equation, since $[x^k]$ interacts with L and E in a nice way. We have

$$[x^0]L^{-1} = [x^0] - [x^0]c[x^0] = [x^0] - c[x^0] = \overline{a}[x^0]\,, \tag{2.21a}$$

$$[x^0]E = [x^0] - [x^0]\overline{x}[\overline{x}] = [x^0]\,. \tag{2.21b}$$

When we extract the x^0 term from the functional equation, we have

$$[x^0](L^{-1} - tES)Q = (\overline{a}[x^0] - t[x^0]S)Q = [x^0](\overline{a} - tS)Q\,. \tag{2.22}$$

Therefore, we have a functional equation for the first coefficient:

$$[x^0](1 - atS)Q = 1\,. \tag{2.23}$$

We can perform the extraction immediately and it simplifies to

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,. \tag{2.24}$$

Extracting higher coefficients is not difficult either. If $k \geqslant 1$, we have

$$[x^k]L^{-1} = [x^k] - [x^k]c[x^0] = [x^k]\,, \tag{2.25a}$$

$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]\,. \tag{2.25b}$$

When we apply the extraction on the functional equation we obtain

$$[x^k](1 - tS)Q = 0, \tag{2.26}$$

and after we expand and simplify the expression, we have

$$(1 - S_0 t)Q_k - t(S_{-1}Q_{k+1} + S_1 Q_{k-1}) = 0\,. \tag{2.27}$$

Proposition 2.3. *Let $S = S_{-1}\overline{x} + S_0 + S_1 x$ be the Laurent step polynomial for a one-dimensional lattice walk model. Then the function Q_0 and Q_k for $k \geqslant 1$ satisfy the equations*

$$(1 - atS_0)Q_0 - atS_{-1}Q_1 = 1\,, \tag{2.28a}$$

$$(1 - S_0 t)Q_k - t(S_{-1}Q_{k+1} + S_1 Q_{k-1}) = 0\,. \tag{2.28b}$$

2.2 Solving Functional Equations For Various Step Sets

If some steps in the step set of a model cannot be taken by walks, then we can remove them and obtain a simpler model with the same lattice walks and generating function. For one-dimensional models with small steps we can identify three major cases: $S = \delta$ or $S = \delta + x$ or $S = \delta + \overline{x} + x$.

2.2.1 The $S = \delta$ or $S = \delta + x$ case

The condition allows us to find Q_0 directly from the $[x^0]$ equation. Then

$$Q_0 = \frac{1}{1 - a\delta t} \tag{2.29}$$

and we can substitute it into the master functional equation to derive the following result.

Proposition 2.4. *Let $S = \delta$ or $S = \delta + x$. Then the generating function is*

$$Q(x, a; t) = \frac{\overline{a}}{1 - tS}\left(1 + \frac{b}{1 - a\delta t}\right) = \frac{1 - \delta t}{(1 - a\delta t)(1 - tS)}\,. \tag{2.30}$$

Something interesting starts to happen once we consider the relationship between $Q(x, a; t)$ and

$\widehat{Q} = Q(x, 1; t)$, its version without interactions. We compute that

$$\widehat{Q} = \frac{1}{(1 - tS)} \tag{2.31}$$

and we have the linear relation

$$(1 - a\delta t)Q - (1 - \delta t)\widehat{Q} = 0\,, \tag{2.32}$$

that we can rewrite as $B((1 - a\delta t)Q) = 0$ with the operator $B = 1 - [b^0]$. The same relationship holds for Q_0.

Proposition 2.5. *Let $B = 1 - [b^0]$ and let Q be the generating function for a lattice walk model with $S = \delta$ or $S = \delta + x$. Then Q and Q_0 both satisfy the equation $B((1 - a\delta t)F) = 0$.*

We will find that similar relationships exist for higher-dimensional lattice walk models too.

2.2.2 The $S = \delta + \overline{x} + x$ case

To derive the generating function in this case, we will use the obstinate kernel method as detailed in Section 4.2 of [30, p. 12] and Section 4.1 of [21, p. 13] for two-dimensional models. In this method we substitute an x value in the functional equation that eliminates the term $(1-tS)Q$. It is sufficient to find roots of the kernel $K = 1 - tS$. The equation $K = 0$ is quadratic in x and has the roots

$$\sigma = \frac{1 - \delta t - \sqrt{(1 - \delta t)^2 - 4t^2}}{2t} \quad \text{and} \quad \sigma^* = \frac{1 - \delta t + \sqrt{(1 - \delta t)^2 - 4t^2}}{2t}\,, \tag{2.33}$$

which multiply to 1, so $\overline{\sigma} = \sigma^*$. Assuming that the substitution $x = \sigma$ eliminates KQ, we find

$$-(c - t\overline{\sigma})Q_0 = \overline{a} \tag{2.34}$$

and using this relation, we can solve for Q explicitly. We compute

$$Q = \frac{\overline{a}}{1 - tS}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - tS}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)\,. \tag{2.35}$$

Proposition 2.6. *Let $S = \delta + \overline{x} + x$. Then the generating functions for this model are*

$$Q_0 = \frac{1}{a\overline{\sigma}t - b} \quad \textit{and} \quad Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}\,, \tag{2.36}$$

where $\overline{\sigma} = (1 - \delta t + \sqrt{(1 - \delta t)^2 - 4t^2})/(2t)$.

As before, we can find a linear relationship between functions for models with interactions and ones without. We have the following result.

Proposition 2.7. *Let $S = \delta + \overline{x} + x$. Then Q and Q_0 satisfy the equation $B((a\overline{\sigma}t - b)F)$.*

In this case we may use a different method, the algebraic kernel method, to obtain a linear relation for Q_0. This method is used extensively in the quarter plane model classification [8]. Here we will find a polynomial $f(a,t)$, for which $B(fQ_0) = c$, instead of zero. Note that K is invariant under the map $* : x \mapsto \overline{x}$ and we can write a second functional equation to complement the first. That is, we have the system of equations

$$(1 - tS)Q - (c - t\overline{x})Q_0 = \overline{a}\,, \tag{2.37}$$

$$(1 - tS)Q^* - (c - tx)Q_0 = \overline{a}\,. \tag{2.38}$$

To eliminate the Q_0 term ,we multiply the first equation by $c - tx$, the second by $c - t\overline{x}$, and subtract one equation from the other. We divide everything by $1 - tS$ and the equation that remains is

$$(c - tx)Q(x) - (c - t\overline{x})Q(\overline{x}) = \frac{\overline{a}t(\overline{x} - x)}{1 - tS} = \overline{a}F\,, \tag{2.39}$$

and crucially, F does not depend on a. We extract the x^1 coefficient from the equation to find an equation relating Q_1 and Q_0

$$cQ_1 - tQ_0 = \overline{a}F_1\,. \tag{2.40}$$

If we set $a = 1$ here, we find that $F_1 = -t\widehat{Q}_0$ and we can substitute that into the previous equation. After we multiply both sides by a, we find that

$$bQ_1 = t(aQ_0 - \widehat{Q}_0)\,. \tag{2.41}$$

Earlier we had the equation

$$(1 - a\delta t)Q_0 = 1 + atQ_1\,, \tag{2.42}$$

which we can solve for Q_1, and insert the solution into the equation above. That is,

$$b((1 - a\delta t)Q_0 - 1) = t^2 a(aQ_0 - \widehat{Q}_0)\,. \tag{2.43}$$

Collecting coefficients and simplifying leads us to the equation

$$(c(1-a\delta t)-at^2)Q_0-(-t^2)\widehat{Q}_0 = c\,, \tag{2.44}$$

and therefore, $f(a,t)=c(1-a\delta t)-at^2$.

Proposition 2.8. *Let $S=\delta+\overline{x}+x$. Then $B((c(1-a\delta t)-at^2)Q_0)=c$.*

We obtained slightly different relationships upon applying the operator B.

2.2.3 Conclusion

In the one-dimensional case we were able to find explicit solutions for Q_0 and Q and various relationships for Q and $\widehat{Q}$, using the operator B. In higher dimensions the analysis will be more difficult, but sometimes we will see the same situation. In some more difficult examples we will only find a relationship between the function for walks returning to the origin and its various partially-marked versions. Here is a table showing the results for the one-dimensional lattice walk models with small steps and interaction with the origin.

S	Q_0	Q	$F(a,t)$
δ	$\dfrac{1}{1-a\delta t}$	$\dfrac{1}{1-a\delta t}$	$1-a\delta t$
$\delta+x$	$\dfrac{1}{1-a\delta t}$	$\dfrac{1-\delta t}{(1-t(\delta+x))(1-a\delta t)}$	$1-a\delta t$
$\delta+\overline{x}+x$	$\dfrac{1}{a\overline{\sigma}t-b}$	$\dfrac{t(\overline{\sigma}-\overline{x})}{(1-t(\delta+\overline{x}+x))(a\overline{\sigma}t-b)}$	$a\overline{\sigma}t-b$

Table 2.1: Generating functions for walks with small steps on the non-negative half-line, where $B(FQ_0)=B(FQ)=0$.

Chapter 3

Higher-dimensional Lattice Walk Models

We begin the exploration of higher-dimensional models by making some useful definitions. Here are the main ingredients for a model set in $\mathbb{Z}^d$.

- A fixed finite set $\mathcal{S}$ of steps for travel between points in the space. It may involve arbitrarily-many finitely-sized steps with multiplicities at least one. Its corresponding Laurent polynomial is denoted by S.
- A region $\mathcal{D} \subseteq \mathbb{Z}^d$, where the walks are free to grow but not venture outside of the boundaries. This may be the whole space, an orthant, a semi-infinite strip, finite boxes, and so on.
- A subregion $\mathcal{D}_0$ of $\mathcal{D}$, where interactions take place. To each point p there will be a corresponding label a_p whose power we can increment upon visits to p. Common subregions $\mathcal{D}_0$ include the origin, the axes in two dimensions, various higher-dimensional hyperplanes, a cone, an infinite strip, the sides of a box and so on.
- A set $\mathcal{Q}$ of initial walks that we can evolve. It has a corresponding generating polynomial q.

We will be growing lattice walks from an initial condition $\mathcal{Q}$, with steps $\mathcal{S}$, within $\mathcal{D}$ in $\mathbb{Z}^d$, while counting interactions within $\mathcal{D}_0$. The four sets determine the model completely.

Definition 3.1. *A lattice walk model $\mathcal{M}$ is characterized by the following four parts: $\mathcal{S}$, $\mathcal{D}$, $\mathcal{D}_0$, $\mathcal{Q}$.*

Definition 3.2. *Define the classic d-dimensional lattice walk model to be the model with small steps in the positive orthant in $\mathbb{Z}^d$. The interacting classic model is the classic d-dimensional model with interactions with the hyperplanes $x_k = 0$ bounding the non-negative orthant.*

Example 3.3. Consider one of the one-dimensional models from the previous chapter. This is an example of a an interacting classic one-dimensional model. Here $\mathcal{D}$ is the set of non-negative integers, $\mathcal{D}_0$ consists of the point at the origin, where the marking variable is a, $\mathcal{Q}$ consists of the empty walk. Finally, $\mathcal{S}$ is one of the subsets of $\{\text{-}1, 0, 1\}$.

Example 3.4. Consider classic models with interactions in two dimensions. Here $\mathcal{D} = \{(x_1, x_2) \in \mathbb{Z}^2 \mid x_1,\, x_2 \geqslant 0\}$ and $\mathcal{D}_0$ consists of marking points on the non-negative axes — x_1 with label a_2, since $x_2 = 0$ here, and x_2 with label a_1, since $x_1 = 0$. Note that the origin is marked with both labels. As in the previous example, $\mathcal{Q}$ consists of the empty walk and $\mathcal{S}$ is some set of small steps, where each step has coordinates of size at most 1.

We also need to keep track of all the properties of the lattice walks in order to enumerate them, so we will use a generating function that tracks quantities like length and ending position. Since the sets $\mathcal{S}$ and $\mathcal{Q}$ are finite, the number of walks of any given length will be finite as well. Therefore, the coefficients of Q will be Laurent polynomials.

Definition 3.5. *Let $\mathcal{M}$ be a lattice walk model and Q its associated generating function counting lattice walks' lengths with t, ending locations with $x_1, \ldots, x_d$, and interactions with $\mathcal{D}_0$ with some multi-index variable a. Then we write*

$$Q = Q(x_1, \ldots, x_d, a; t) = \sum_{k=0}^{\infty} q_k(x_1, \ldots, x_d, a) t^k \,, \tag{3.1}$$

where the coefficients q_k are Laurent polynomials in the x and a variables.

Example 3.6. In this example we will examine the simplest kind of model — one where $\mathcal{D}$ is the whole space, there are no interactions and $\mathcal{Q}$ consists of the empty walk. Since the lattice walks may pick any step from the step set $\mathcal{S}$, the walks of length one are enumerated via tS, the walks of length two, via t^2S^2, and so on, where $S = S(x_1, \ldots, x_d)$. Then the full generating function is

$$Q(x_1, \ldots, x_d; t) = 1 + St + S^2t^2 + \cdots = \frac{1}{1 - St} \,. \tag{3.2}$$

Example 3.7. Consider the two-dimensional classic model with $\mathcal{S} = \{\text{NE, NW, SE, SW}\}$, that is, $S = (\overline{x}_1 + x_1)(\overline{x}_2 + x_2)$. The only walk of length one is enumerated by x_1x_2t. When we grow the walk, it may take any of the four available steps and we have

$$Q = 1 + x_1x_2t + (a_1 + x_1)(a_2 + x_2)t^2 + \cdots \,. \tag{3.3}$$

We may extract coefficients too. For example, the walks ending on the x_1 axis are given by setting $x_2 = 0$ or extracting the x_2^0 term. That is,

$$[x_2^0]Q = 1 + a_2(a_1 + x_1)t^2 + \cdots . \tag{3.4}$$

The walks returning to the origin are given by

$$[x_1^0 x_2^0]Q = 1 + a_1 a_2 t^2 + \cdots . \tag{3.5}$$

While it is possible to handle the analysis of higher-dimensional classic models without extra notation, it will greatly help us to have it. In fact, notation conventions paved the way to finding the functional equations for higher-dimensional models and later, the operators. With this thesis, we aim to make this kind of notation more common in the enumeration of lattice walks. Typically, in the multivariate setting we see multi-index notation where a bolded or vectorial form denotes the collection of variables involved, but such notation is not enough when working with restricted geometries for regions and marking subregions. We have the following notation.

Definition 3.8. *Let $\mathbb{0} := (0, \ldots, 0)$ and $\pm\mathbb{1} := (\pm 1, \ldots, \pm 1)$, where we have as many terms in these sequences as needed in the specific context. Also, let $[n] = \{1, \ldots, n\}$.*

Definition 3.9. *Let $x_1, \ldots, x_d$ be indeterminates and let $I = (i_1, \ldots, i_m)$ and $J = (j_1, \ldots, j_m)$ be sequences of positive integers. Define the monomial x_I^J as*

$$x_I^J = x_{i_1}^{j_1} \cdots x_{i_m}^{j_m} . \tag{3.6}$$

Then let the coefficient and term extraction operators be defined as

$$[x_I^J]F = [x_{i_1}^{j_1} \cdots x_{i_m}^{j_m}]F = F_I^J \quad \textit{and} \quad [[x_I^J]] = x_I^J [x_I^J] . \tag{3.7}$$

For the monomial, we have the special cases $x_I^{\mathbb{0}} = 1$ and $x_I^{\mathbb{1}} = x_I$ and $x_I^{-\mathbb{1}} = 1/x_I = \overline{x}_I$. For the coefficient extraction, we have the special cases $F_I^J = F$ if I is empty, and $F_I^J = F_J$ if I is the full set of variables.

Example 3.10. In the previous example we had $[x_2^0]Q = Q_2^0$ and $[x_1^0 x_2^0]Q = Q_{1,2}^{0,0}$. Furthermore, we have $S_{1,2}^{\pm 1, \pm 1} = 1$ and

$$S_1^{-1} = [\overline{x}_1](x_1 + \overline{x}_1)(x_2 + \overline{x}_2) = x_2 + \overline{x}_2 . \tag{3.8}$$

Because there are different variables involved, there may be some ambiguity in the result, depending on the order of extraction. We will always perform coefficient extractions with respect to the x or a variables by first expanding any series in t, and then applying the extraction on the coefficients. The following notation makes this precise.

Definition 3.11. *Suppose that Γ is an operator involving x and a variables but not t. We apply Γ on power series $F = \sum_{k=0}^{\infty} f_k t^k$ in the following way:*

$$\Gamma F = \sum_{k=0}^{\infty} \Gamma(f_k) t^k \,. \tag{3.9}$$

Example 3.12. In the past we have performed inclusion-exclusion to arrive at functional equations for the two-dimensional problems. This is the usual procedure carried out in problems so far, like in [8], but when it comes to generalizing to models with interactions, the process is difficult and prone to error. For a view of difficulties involved in not using the operator notation, see [30], where the authors had to wrestle with case-by-case analysis. Instead, we use operators to derive the functional equations cleanly. As in the one-dimensional case with generating function $Q = \sum_{k=0}^{\infty} q_k t^k$, we may grow the set of walks by performing the multiplication Sq_k. Then we assume that an operator Γ exists such that it cleans up the terms in Sq_k. We obtain the functional equation $(1 - t\Gamma S)Q = 1$ again and all that is left to be determined is the operator Γ itself. We can make and observations that will allow us to find it easily. The one-dimensional operator $1-[[\overline{x}]]$ is idempotent, that is $E^2 = E$, and combinatorially-speaking, once we remove certain terms, we do not remove them again. In two dimensions this means that with the use of one-dimensional operators, we may remove terms with negative x_1 coordinate first, and ones with negative x_2 coordinate second, without performing inclusion-exclusion. That is, the operator E here is given by $E_1 E_2$ where $E_k = 1 - [[\overline{x}_k]]$. Multiplying these out, we find that

$$E = 1 - [[\overline{x}_1]] - [[\overline{x}_2] + [[\overline{x}_1 \overline{x}_2]] \tag{3.10}$$

and the operator iteslf encodes the inclusion-exclusion. The situation is analogous for the marking operator $L = L_1 L_2$, where $L_k = 1 + b_k[x_k^0]$. Again, the operator L is invertible and analogously, we may write the functional equation as $(L^{-1} - tES)Q = \overline{a}_1 \overline{a}_2$. If we perform the coefficient extractions, we will find the equation

$$Q = 1 + tSQ - t\overline{x}_1 S_1^{-1} Q_1^0 - t\overline{x}_2 S_2^{-2} Q_2^0 + t\overline{x}_1 \overline{x}_2 S_{1,2}^{-1,-1} Q_{1,2}^{0,0} \,. \tag{3.11}$$

Contrast this notation to the one in [8, p. 10]. The problem quickly runs into notational difficulties for d-dimensional problems, but that can be dealt with by writing the equation as a sum over subsets of $\{1, \ldots, d\}$, like it is done for the non-interacting case in Equation 5.1 in [15, p. 58]. This method runs into trouble when we start considering interactions. We take the functional equation for the non-interacting case and modify it to add interactions. This analysis has been done before in Section 3 of [30, p. 7] and Section 3.2 of [21, p. 11], where the authors add in terms like $(a_1 - 1)T$, where T is some expression involving extractions of Q and S, but we will not go into the messy details. Instead, we will use the operator notation to handle the problem in a more efficient manner.

3.1 Using Operators to Derive the Functional Equation

As we have seen in the previous example, it is not difficult to generalize the derivation of the functional equations if we use operators. The derivation of general case for d dimensions is analogous to the one in the example, so we will omit it most of it. In d dimensions, the interactions take place on the hyperplanes where $x_k = 0$, so we will use the labels a_k to mark those interactions. The operators here are $L = L_1 \cdots L_d$ and $E = E_1 \cdots E_d$, where order does not matter. We need to compute the quantity $L^{-1}(1)$, so we have

$$L^{-1}(1) = \sum_{I \subseteq [d]} (-1)^{|I|} c_I = \overline{a}_{[d]} \,. \tag{3.12}$$

The higher-dimensional functional equation for the interacting classic model is given as follows.

Proposition 3.13. *The classic interacting d-dimensional lattice walk model has a full generating function Q satisfying the functional equations*

$$(1 - tLES)Q = 1 \,, \tag{3.13}$$

$$(L^{-1} - tES)Q = \overline{a}_{[d]} \,, \tag{3.14}$$

where L, E, L^{-1} are of the form $\Gamma = \Gamma_1 \cdots \Gamma_d$.

Previously, we have had to cope with coefficient extractions and back-substitution to find the expanded form of the second functional equation [30, p. 9] and [21, p. 12]). We can skip this process by using operators. In general, the second functional equation has many less terms. To see this, we take a look at the operator $L_k E_k$. We have

$$L_k E_k = 1 + b_k[x_k^0] - \overline{x}_k[\overline{x}_k] \tag{3.15}$$

and if we apply the operator to an equation, we increases the number of terms in the equation by a factor of 3. We have 3^d terms relating to SQ by the end of applying LE. On the other hand, once we apply the operator L^{-1}, we introduce 2^d terms relating to Q, but since we only have E applied on SQ, we would only have at most 2^d terms. We have exactly 2^d terms $Q_I^{\mathbb{0}}$, one for each subset $I \subseteq [d]$. In this case, we can write out the functional equation explicitly. First,

$$L^{-1}Q = (1 - c_1[x_1^0]) \cdots (1 - c_d[x_d^0])Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{0}}\,, \tag{3.16a}$$

$$ESQ = \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [\overline{x}_I](SQ) = \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}}\,, \tag{3.16b}$$

where the last equality follows from the fact that there are no negative powers of x_k in Q for any k. Then the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x}_I S_I^{-\mathbb{1}}) Q_I^{\mathbb{0}} = \overline{a}_{[d]}\,. \tag{3.17}$$

We let $K_I = c_I - t\overline{x}_I S_I^{-\mathbb{1}}$, where, if $I = \varnothing$, we have the empty products and empty extractions, leaving us with $K_\varnothing = K$. Finally, we can write the master equation in d dimensions.

Theorem 3.14. *For subsets I of $\{1, \ldots, d\}$, let $K_I = c_I - t\overline{x}_I S_I^{-\mathbb{1}}$. Then the classic interacting d-dimensional lattice walk model has a generating function satisfying*

$$\sum_{I \subseteq [d]} (-1)^{|I|} K_I Q_I^{\mathbb{0}} = \overline{a}_{[d]}\,. \tag{3.18}$$

Equation 5.1 of [15, p. 58] is the non-interacting version of our equation, and it can be written as

$$KQ - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{1}} Q_I^{\mathbb{0}} = 1\,. \tag{3.19}$$

3.2 Coefficient Extractions

In this section we will see how the extraction and marking operators interact with basic coefficient extraction to produce quicker coefficient extraction of the functional equation. That is, it is faster to use the operator form of the functional equation in order to extract coefficients than it is to use the explicit form. However, this section is not essential for our understanding of the enumeration of lattice walk models. The main role of the following analysis is to provide a shortcut for manual calculations. After all, we need to obtain functional equations for $Q_I^{\mathbb{0}}$ for $I \subseteq [d]$ like in the one-

dimensional case.

3.2.1 Using Operator Notation For Fast Coefficient Extraction

We will find the $x_I^{\mathbb{0}}$ term of the functional equation to obtain further equations for the functions $Q_I^{\mathbb{0}}$. The operator $[x_i^k]$ commutes with operators $M_j = L^{-1}$ and E_j when $j \neq i$, so we only need to consider the action on M_i and E_i. Similarly to Equation 2.21 and Equation 2.25, in the two-dimensional setting we have

$$[x_i^k]M_i = \overline{a}_i^{\delta(k,0)}[x_i^k] \quad \text{and} \quad [x_i^k]E_i = [x_i^k]\,. \tag{3.20}$$

Therefore, for $k \geqslant 1$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture, while for $k = 0$ it removes E_i but leaves a residual $\overline{a}_i$ behind for M_i. Therefore, if we extract $[x_I^{\mathbb{0}}]$ from the functional equation for $I \subseteq [d]$, we have

$$\begin{aligned}
[x_I^{\mathbb{0}}]a_{[d]} = \overline{a}_{[d]} &= [x_I^{\mathbb{0}}](M_{[d]} - tE_{[d]}S)Q \\
&= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{0}}] - tE_{[d]\setminus I}[x_I^{\mathbb{0}}]S\right)Q \\
&= [x_I^{\mathbb{0}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q\,,
\end{aligned} \tag{3.21}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{0}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q\,. \tag{3.22}$$

When we extract a general term, we use the extraction $[x_I^{\mathbb{0}}x_J^P]$, where P is a nonempty set of integers greater than zero and $I \cup J = K$ with no overlaps. Through the same process as before, we find the equation

$$[x_I^{\mathbb{0}}x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0\,. \tag{3.23}$$

3.2.2 Further Functional Equations

If we want to find equations similar to the master functional equation, we will need to perform the extraction explicitly like in Equation 3.16. First,

$$[x_I^{\mathbb{0}}]SQ = S_I^{\mathbb{0}}Q_I^{\mathbb{0}} + R_I\,, \tag{3.24}$$

where R_I is a collection of $S_I^{-P}Q_I^P$ terms that involve sets P with some nonzero entries. We can use this term inside the functional equation as

$$\begin{aligned}\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{0}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{0}}](SQ) \\ &= M_{[d]\setminus I}Q_I^{\mathbb{0}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{0}}Q_I^{\mathbb{0}} + R_I) \\ &= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{0}})Q_I^{\mathbb{0}} - a_I t E_{[d]\setminus I}R_I\end{aligned} \tag{3.25}$$

so we can rewrite it as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I}S_I^{\mathbb{0}})Q_I^{\mathbb{0}} = a_I(\overline{a}_{[d]} + tR_I')\,, \tag{3.26}$$

where $R_I' = E_{[d]\setminus I}R_I$. We compute the following terms

$$M_{[d]\setminus I}Q_I^{\mathbb{0}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}c_J Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.27a}$$

$$a_I t E_{[d]\setminus I}S_I^{\mathbb{0}}Q_I^{\mathbb{0}} = \sum_{J\subseteq[d]\setminus I}(-1)^{|J|}a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}}Q_{I,J}^{\mathbb{0},\mathbb{0}}\,, \tag{3.27b}$$

and the functional equation can be written as

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}(c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}})Q_{I,J}^{\mathbb{0},\mathbb{0}} = a_I(\overline{a}_{[d]} + tR_I')\,. \tag{3.28}$$

The form of this expression suggests the following notation.

Definition 3.15. *For integer sets I and J that do not share any elements, we write*

$$K_J(I) = c_J - a_I t\overline{x}_J S_{I,J}^{\mathbb{0},-\mathbb{1}}\,, \tag{3.29}$$

where if $I = \varnothing$, then $K_J(I) = K_J$.

With this notation, we can write the 2^d equations for $Q_I^{\mathbb{0}}$ and $I \subseteq [d]$ in simplified form.

Lemma 3.16. *Let I be a subset of $\{1,\ldots,d\}$. The generating function $Q_I^{\mathbb{0}}$ for the d-dimensional interacting classic lattice walk model satisfies the functional equation*

$$\sum_{J\subseteq[d]\setminus I}(-1)^{|J|}K_J(I)Q_{I,J}^{\mathbb{0},\mathbb{0}} = \overline{a}_{[d]\setminus I} + a_I tR_I'\,, \tag{3.30}$$

where the R_I' term involves only higher-order coefficient extractions from Q.

This is a clean way to write down all the equations for $Q_I^{\mathbb{0}}$ in a unifying way. To compute the R'_I terms, we first determine what R_I is. It is a sum of terms $S_I^{-P}Q_I^P$ for various non-negative P with size $|I|$ whose elements are not all 0. Then we compute $R'_I = E_{[d]\setminus I}R_I$. A key observation is that the terms Q_I^P are power series in the remaining variables, so they do not have non-negative coefficients. Let $sq = S_I^{-P}Q_I^P$. Then

$$E_j sq = sq - \overline{x}_j (sq)_j^{-1} = sq - \overline{x}_j s_j^{-1} q_j^0 \tag{3.31}$$

and thus,

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{1}} q_J^{\mathbb{0}} \,. \tag{3.32}$$

This allows us to derive the following explicit form of R'_I.

Proposition 3.17. *The functions R'_I are given by*

$$R'_I = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{0}} S_{I,J}^{-P,-\mathbb{1}} Q_{I,J}^{P,\mathbb{0}} \,. \tag{3.33}$$

Of course, when $d = 1$ or $d = 2$, we do not need the full generality of the formula, but for higher dimensions, it would be quite difficult to write the equations by hand without it.

3.2.3 Kernel Forms

We can write out the functional equations in a way that highlights the kernel K. For example,

$$(1 - tLES)Q = (1 - t(LE - 1 + 1)S)Q = KQ + t(1 - LE)SQ \,, \tag{3.34}$$

where $1 - LE$ is an operator consisting of coefficient extractions only. Here is another way we could have expanded the operator form.

$$(1 - tLES)Q = LEKQ + (1 - LE)Q = LEKQ + (1 - L)Q \,, \tag{3.35}$$

where $EQ = Q$ since Q has no terms that need removing. Something interesting happens when we set all the a_k's to 1 in the last equation. This is equivalent to removing the marking operator. The equation becomes $E(KQ) = 1$ and this form suggests that we use the following substitution

$$Q = \frac{W}{x_1 \cdots x_d K} \,. \tag{3.36}$$

Then we multiply both sides of the functional equation by $x_1 \cdots x_d$ and obtain

$$x_1 \cdots x_d = x_1 \cdots x_d E(KQ) = x_1 \cdots x_d E(\overline{x}_1 \cdots \overline{x}_d W) = E'W\,, \tag{3.37}$$

where $E' = E'_1 \cdots E'_d$ and

$$E'_k = x_k E_k \overline{x}_k = x_k(1 - \overline{x}_k[\overline{x}_k])\overline{x}_k = 1 - [x_k]^0\,. \tag{3.38}$$

Proposition 3.18. *Let $W = x_1 \cdots x_d KQ$, where Q is the generating function for walks without interactions in the positive orthant. Then*

$$E'W = \sum_{I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{0}} = x_1 \cdots x_d\,. \tag{3.39}$$

For example, in two dimensions we have

$$W(x_1, x_2) - W_1^0(x_2) - W_2^0(x_1) + W_{1,2}^{0,0} = x_1 x_2\,. \tag{3.40}$$

Thus, we can write the solution to the problem as

$$Q = \frac{1}{x_1 \cdots x_d K} \left(x_1 \cdots x_d - \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} W_I^{\mathbb{0}} \right). \tag{3.41}$$

In two dimensions we have

$$Q(x_1, x_2) = \frac{x_1 x_2 + W_1^0(x_2) + W_2^0(x_1) - W_{1,2}^{0,0}}{x_1 x_2 (1 - tS)}\,. \tag{3.42}$$

Example 3.19. Consider the one-dimensional model on the non-negative line with $S = \overline{x} + x$. Here the solution for the interacting version of the model is given by

$$Q = \frac{t(\overline{\sigma} - \overline{x})}{(1 - tS)(a\overline{\sigma}t - b)}, \tag{3.43}$$

so when we set $a = 1$ and simplify the expression, we find that the generating function for the non-interacting model is given by

$$\widehat{Q} = \frac{x - \sigma}{x(1 - tS)}\,, \tag{3.44}$$

as predicted by the analysis of this section. Here $W_0 = \sigma$.

Of course, each $W_I^{\mathbb{0}}$ depends on $Q_I^{\mathbb{0}}$, so we could have seen this relationship starting with the master functional equation for Q itself. Interestingly enough, the calculation points to a differential equation. Clearly, W satisfies the differential equation $\partial_1\partial_2 W = 1$ and a similar relationship exists for the d-dimensional models.

Proposition 3.20. *The generating function for a d-dimensional classic model, satisfies the PDE*

$$\partial_1 \cdots \partial_d (x_1 \cdots x_d K Q) = 1\,. \tag{3.45}$$

The result is not difficult to generalize. If we have walks with larger steps, we will have a similar partial differential equation where the orders of variables and orders of the derivatives will be higher. If we have interactions, we will also have higher order terms.

3.3 First Solutions

3.3.1 Formal Neumann Series

So far we have obtained the equation $(1 - t\Gamma S)Q = 1$, where $\Gamma = LE$. Formally, we can expand this operator as Neumann series and express the generating functions as

$$Q = \sum_{k=0}^{\infty} t^k (\Gamma S)^k(1)\,, \tag{3.46}$$

where $(\Gamma S)^k(1)$ indicates repeated application of ΓS. This allows us to compute Q iteratively, where the next t-coefficient is the previous one with ΓS applied to it. This is a formalization of the naive brute force approach of growing lattice walks from smaller ones: by taking a set of walks of length n, extending them by 1 step, removing the ones that end outside of the quarter plane, and finally by marking the ones that interact with the hyperplanes $x_i = 0$ at their endpoint. We have not been able to find closed forms for the repeated operator applications in general. Here is an example that illustrates the use of the operators.

Example 3.21. Let $S = x_1 + \overline{x}_1 + x_2 + \overline{x}_2 = S_1(x_1) + S_2(x_2)$. We start with $(\Gamma S)^0(1) = 1$. Then

$$(\Gamma S)(1) = LE(S_1 + S_2) = L(x_1 + x_2) = a_2 x_1 + a_1 x_2\,, \tag{3.47}$$

$$\begin{aligned}(\Gamma S)^2(1) &= LE(S_1 + S_2)(a_2 x_1 + a_1 x_2) \\ &= LE(a_2 S_1 x_1 + a_1 x_2 S_2 + a_2 x_1 S_2 + a_1 x_2 S_1) \\ &= a_1 a_2^2 + a_2^2 x_1^2 + a_1 a_2^2 + a_1^2 x_2^2 + (a_1 + a_2) x_1 x_2\,.\end{aligned} \tag{3.48}$$

3.3.2 Hadamard Products

We take a moment to see a result that falls out of the operator notation. In Section 5 of [18, p. 14], the authors consider the special form of the step polynomial $S(x_1, x_2) = S_0(x_1) + S_1(x_1)S_2(x_2)$ for the non-interacting case. Here we will find a similar result for the simpler case $S = S_1(x_1)S_2(x_2)$ but with interactions, and its generalizations to arbitrarily-many dimensions.

Example 3.22. When $S = S_1(x_1)S_2(x_2)$ the operator ΓS factors as

$$\Gamma S = \Gamma_1\Gamma_2 S_1 S_2 = (\Gamma_1 S_1)(\Gamma_2 S_2)\,. \tag{3.49}$$

This is because Γ_2 does not interact with S_1 and the operators commute. Then

$$(\Gamma S)^k = (\Gamma_1\Gamma_2 S_1 S_2)^k = (\Gamma_1 S_1)^k(\Gamma_2 S_2)^k \tag{3.50}$$

and we use the Neumann series to find

$$Q = \sum_{k=0}^{\infty} t^k(\Gamma S)^k = \sum_{k=0}^{\infty} t^k(\Gamma_1 S_1)^k(\Gamma_2 S_2)^k = \left(\sum_{k=0}^{\infty} t^k(\Gamma_1 S_1)^k\right) \odot \left(\sum_{k=0}^{\infty} t^k(\Gamma_2 S_2)^k\right)\,, \tag{3.51}$$

where $\odot$ denotes the Hadamard product of two power series with respect to variable t. By using

$$Q_i(a_i, x_i; t) = \sum_{k=0}^{\infty} t^k(\Gamma_i S_i)^k \tag{3.52}$$

and reversing the iteration, we reach the functional equations $(1 - t\Gamma_i S_i)Q_i = 1$ for each i. The functions Q_i are solutions to one-dimensional problems with step sets S_i and Q is expressed as

$$Q(a_1, a_2, x_1, x_2; t) = Q_1(a_1, x_1; t) \odot Q_2(a_2, x_2; t) \tag{3.53}$$

Example 3.23. Once we know the form of the solution, we can prove that it works by a method that does not use the series solution. Suppose that Q_i is the solution to the functional equation

$$(L_i^{-1} - tE_i S_i)Q_i = \overline{a}_i \tag{3.54}$$

for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional equation

$$(L^{-1} - tES)Q = \overline{a}_1\overline{a}_2\,. \tag{3.55}$$

To this end, we compute

$$\begin{aligned}(L^{-1} - tES)Q &= L^{-1}(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) \\ &= (L_1^{-1}Q_1 \odot L_2^{-1}Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2))\,.\end{aligned} \tag{3.56}$$

Then we substitute $L_i^{-1}Q_i = \overline{a}_i + tE_iS_iQ_i$ into the first term and expand it to find

$$\begin{aligned}(L_1^{-1}Q_1 \odot L_2^{-1}Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) \\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) \\ &\quad + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\,.\end{aligned} \tag{3.57}$$

Since we are working with power series,

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0 \tag{3.58}$$

and we are done.

Lemma 3.24. *Given an interacting classic model, with $S = S_1(x_1)S_2(x_2)$, the function Q satisfies*

$$Q(x_1, x_2, a_1, a_2; t) = Q_1(x_1, a_1; t) \odot Q_2(x_2, a_2; t)\,, \tag{3.59}$$

where Q_i is the one-dimensional generating function for the model with for step polynomial S_i.

This identity will serve as the base case for the following general case.

Theorem 3.25. *Given an interacting classic model with $S = S_1 \cdots S_r$, where S_i and S_j do not share variables when $i \neq j$, let Q and Q_i be the generating functions for models with step polynomials S and S_i, respectively. Then $Q = Q_1 \odot \cdots \odot Q_r$.*

Example 3.26. Consider the two-dimensional model with diagonal steps, that is, $S = (x_1 + \overline{x}_1)(x_2 + \overline{x}_2)$. We will look at this case in more detail in Chapter 5. The one-dimensional problems are the same and have solutions

$$\widetilde{Q}_0 = \frac{\sigma}{at - b\sigma} \quad \text{and} \quad \widetilde{Q} = \frac{t(x - \sigma)}{x(1 - t(x + \overline{x})(at - b\sigma)}\,, \tag{3.60}$$

where

$$\sigma = \frac{1 - \sqrt{1 - 4t^2}}{2t}\,. \tag{3.61}$$

We can expand and simplify $\widetilde{Q}_0$ and $\widetilde{Q}$ to find

$$\widetilde{Q}_0 = \frac{2 - a - a\sqrt{1-4t^2}}{2(1-a+a^2t^2)}\,, \tag{3.62}$$

$$\widetilde{Q} = \frac{((a-2)t + (1-a)x + 2axt^2) + (at + (1-a)x)\sqrt{1-4t^2}}{2(1-a+a^2t^2)(x-t-tx^2)}\,. \tag{3.63}$$

We have the expansion

$$\widetilde{Q} = 1 + xt + (a + x^2)t^2 + x(1 + a + x^2)t^3 + \cdots \tag{3.64}$$

and finally,

$$\begin{aligned} Q(x_1, x_2, a_1, a_2; t) &= \widetilde{Q}(x_1, a_1; t) \odot \widetilde{Q}(x_1, a_2; t) \\ &= 1 + x_1x_2t + (a_1 + x_1^2)(a_2 + x_2^2)t^2 \\ &\quad + x_1x_2(1 + a_1 + x_1^2)(1 + a_2 + x_2^2)t^3 + \cdots\,. \end{aligned} \tag{3.65}$$

However, obtaining an explicit function via the Hadamard product seems to be rather messy. The function is not algebraic since the unmarked variant is not algebraic [8, p. 26]. However, the function Q is D-finite since both functions in the product are.

3.3.3 Reducible Models and Groups For Walks With Small Steps

Here is another observation that will allow us to find solutions to models in an easier way. As observed in [8, p.5], if the model has some steps that cannot be used in the construction of lattice walks, then we can safely remove them from the step set, and we would still have the same generating function enumerating the lattice walks. This procedure works for any step set and any regional constraints, so we make a definition.

Definition 3.27. *For a model with a step set $\mathcal{S}$, let the minimal step set $\mathcal{S}'$ for be the set of steps that can eventually be used by lattice walks. A model is reducible if $\mathcal{S}'$ is a proper subset of S.*

Lemma 3.28. *Whether we use $\mathcal{S}$ or $\mathcal{S}'$, the full generating function for the model is the same.*

Next, it is easier to find the generating function for a model if we have more functional equations because this may help us cancel out some of the terms. One way to find more equations is to apply a transformation on the tuple $(x_1, \ldots, x_d)$ that leaves intact as many parts of the functional equation as possible but still adds some information in the form of a non-equivalent equation. We could map each $x_1, \ldots, x_d$ to some functions separately and then examine the group generated by such maps. We often eliminate the kernel in front of Q in order to find equations for the extracted forms of

Q, so it is useful to consider transformations that preserve the kernel. This is a fruitful approach in the non-interacting case, and for its application on two-dimensional models, see Section 4.2 [8, p. 11]. We will call this group $G(K)$, although it is not the full group of rational transformations preserving the kernel. For example, a lot of the time $(x_1, x_2) \mapsto (x_2, x_1)$ is not in the group, but it is a valid transformation preserving the kernel. If a transformation on $(x_1, \ldots, x_d)$ preserves the kernel $K = 1 - tS$, then it must also preserve the step polynomial S. We may write S as

$$S = S_k^{-1}\overline{x}_k + S_k^0 + S_k^1 x_k \tag{3.66}$$

for some k from 1 to d. We send $x_k \mapsto y_k$ and solve $S(x_k) = S(y_k)$ for y_k to determine the possible transformations we can perform. After some simplification, we obtain the equation

$$(S_k^{-1} - S_k^1 x_k y_k)(y_k - x_k) = 0\,. \tag{3.67}$$

We have three cases depending on which ones of S_k^{-1} and S_k^1 are zero.

1. If both are zero, the equation is trivially-true and all transformations preserve the kernel.

2. If only one of them is zero, then $y_k = x_k$, so only the identity transformation preserves the kernel.

3. If neither one is zero, we have $y_k = x_k$ or y_k is the involution $y_k = \overline{x}_k S_k^{-1}/S_k^1$.

Case one and two are mostly trivial and we find properties of the generating function right away.

1. If $S_k^{-1} = 0$ and $S_k^1 = 0$, then $S = S_k^0$ and the model takes all of its steps inside the $x_k = 0$ hyperplane. Therefore, the model is equivalent to a lower-dimensional model with no x_k coordinate. Suppose that the generating function in the latter case is $q(t)$. Then the generating function in the former case is given by $Q(x_k, a_k; t) = q(a_k t)$, where the rest of the variables have been suppressed from the notation. We can also find this solution from the functional equation directly. Let $M = M'M_k$ and $E = E'E_k$. Then by Equation 3.22 and the observation that $[x_k^0]SQ = SQ$ and $[x_k^0]Q = Q$, we have

$$(M' - a_k t E' S)Q = \overline{a}_{[d]\setminus\{k\}}\,, \tag{3.68}$$

   This is just an equation for a $(d-1)$-dimensional model with an extra scaling of t by a_k.

2. If $S_k^{-1} \neq 0$ and $S_k^1 = 0$, then the model is reducible to one with $S = S_k^0$, that is, the model reduces to case 1.

3. If $S_k^{-1} = 0$, $S_k^0 = 0$ and $S_k^1 \neq 0$, then the walks in model leave the $x_k = 0$ hyperplane and never return. Here we have an essentially lower-dimensional problem as well, since we have $S = S_k^1 x_k$. If the generating function for the lower-dimensional problem is $q(t)$, then the generating function for this model is $Q(x_k, a_k; t) = q(x_k t)$, where the rest of the variables have been suppressed from the notation. Similarly to case 1, we may derive the result from the functional equation directly.

All the other cases are more complicated and require more careful analysis. We can define a group G_S of the nontrivial transformations.

Definition 3.29. *For a specific step polynomial S, let $G = G_S = G(K)$ be the group generated by the nontrivial transformations g_k.*

It is easy to see that there are infinitely many such groups, just consider $S = \prod_{k=1}^{d}(x_k + \overline{x}_k)$ with $g_k(x_k) = \overline{x}_k$. The group here has order 2^d. The orders of the finite groups set in 2 dimensions are 1, 2, 4, 6 and 8, from Theorem 3 in [8, p. 8]. If we were to consider walks with multiple copies of some steps, we may obtain even larger group orders as seen in Section 7 of [11, p. 11]. The d-dimensional groups defined above are Coxeter groups — a generalization of the dihedral groups encountered in dimension 2 lattice walk models. Coxeter groups are defined as follows.

Definition 3.30. *The group*

$$G = \langle r_1, \ldots, r_d \mid (r_i r_j)^{m_{i,j}} = 1 \rangle \tag{3.69}$$

is known as a Coxeter group. Here $m_{i,i} = 1$ and $m_{i,j} \geqslant 2$ for all $i \neq j$. Also $m_{i,j}$ could be infinite.

We call these r's reflections and when $m_{i,j} = 2$, they commute. Also, $m_{i,j} = m_{j,i}$. The *rank* of G is the number of generators in this specific representation of G. Often G has more than one representation with different numbers of generators. We can draw a graph for each Coxeter group by the following procedure.

- Label vertices with the generators of G.
- Connect the vertices g_i and g_j if $m_{i,j} \geqslant 3$.
- Label the edges with $m_{i,j}$ if $m_{i,j} \geqslant 4$.

If the graph has different components, the Coxeter group factors as a direct product of Coxeter groups, represented by the individual components. We can see this phenomenon in terms of the step polynomial S.

Proposition 3.31. *If the step polynomial S factors into two disjoint step polynomials T and R as*

$$S = T(x_1, \ldots, x_k)R(x_{k+1}, \ldots, x_d)\,, \tag{3.70}$$

then the group G_S factors as

$$G_S = G_T \times G_R\,, \tag{3.71}$$

where G_S, G_T and G_R are the groups for the step polynomials S, T and R, respectively.

Proof. For a fixed integer $1 \leqslant i \leqslant k$,

$$S_i^j = [x_i^j]S = R[x_i^j]T = RT_i^j \tag{3.72}$$

and therefore,

$$g_i(x_i) = \overline{x}_i \frac{T_i^{-1}R}{T_i^1 R} = \overline{x}_i \frac{T_i^{-1}}{T_i^1}\,. \tag{3.73}$$

Similarly, if $k+1 \leqslant j \leqslant d$, then $g_j(x_j) = \overline{x}_j R_i^{-1}/R_i^1$. Furthermore, g_i and g_j commute since they do not affect each other's variables. Since i and j are arbitrary, the generators from different sets commute and the Coxeter graph of the group factors in two, that is, the group itself factors. □

3.4 Specializing to Two Dimensions

Until the end of the chapter we will focus on the two-dimensional models with interactions. We can set $d = 2$ in Equation 3.18 and Equation 3.30 to find the two-dimensional master functional equations.

Corollary 3.32. *The generating function for a two-dimensional classic interacting model satisfies the following functional equations*

$$\begin{aligned}
KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} &= \overline{a}_1\overline{a}_2\,,\\
(1 - ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1t(S_1^{-1}Q_1^1 - \overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) &= \overline{a}_2\,,\\
(1 - ta_2S_2^0)Q_2^0 - (c_1 - ta_2\overline{x}_1S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2t(S_2^{-1}Q_2^1 - \overline{x}_1S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) &= \overline{a}_1\,,\\
(1 - a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) &= 1\,.
\end{aligned} \tag{3.74}$$

3.4.1 First Examples in the Plane

The classic two-dimensional lattice walk models have been investigated thoroughly but the interacting version has not. We want to extend the classic model methods to the interacting setting or create

new ones. Since the generating functions for the interacting models need to reduce to those for the non-interacting models, we do not expect to find solutions for every model. In fact, some of the classic models lead to generating functions that are not even D-finite, so their interacting counterparts will not be D-finite either. For the rest of the problems, we hope to find the generating functions Q explicitly, or at least the functions $Q_{1,2}^{0,0}$. This appears to be difficult in most nontrivial cases. Recent research [30] has shown that there are models, for which we have an explicit relationship between Q or $Q_{1,2}^{0,0}$ and unmarked variants. As we saw in the previous chapter, the one-dimensional cases admit a linear relationship with polynomial coefficients. There are other two-dimensional models that exhibit a rational relationship. The way things have turned with regards to the classic models, it would not be surprising if there are two-dimensional interacting models for which neither a linear nor a rational relationship would exist between Q, $Q_{1,2}^{0,0}$ and various unmarked variants. In this section we will cover the first few examples of quarter plane models with interactions.

Our first step is to find all the reducible models and we can do this by looking at which of the steps $\{$N, NE, E$\}$ they have in their step set.

1. There are 32 models reducible models, where none of the three steps are present and walks cannot leave the origin.



Figure 3.1: Set of steps allowed in models that do not leave the origin. The models are equivalent to a model without steps.

2. There is a $2 \cdot 2^4 = 16$ reducible models with an N step or an E step, where the models are equivalent to one-dimensional ones.

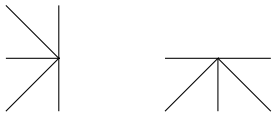


Figure 3.2: Sets of steps allowed in models with an N step or an E step. The models are essentially one-dimensional.

3. There are two or more steps from {N, NE, E} or NE $\subseteq \mathcal{S}$. In this case, the models feature walks that leave both axes and once that happens, such walks may use any step from the step set. Therefore, these models are not reducible.

Cases 1 and 2 contain all of the quarter plane models reducible to one-dimensional models. We will look at the case of a model equivalent to a one-dimensional model along the x_2 axis. From the analysis in the previous section, since both $S_1^{-1} = 0$ and $S_1^1 = 0$, we have the first situation. If the solution to the one-dimensional problem is $q(x, a; t) = q(x_2, a_2; t)$, then the solution to the models in cases 1 and 2 is $Q(x_1, x_2, a_1, a_2; t) = q(x_2, a_2; a_1 t)$. To see it in another way, consider the polynomial S itself in the reduced models. Since S factors, the function Q is a Hadamard product between two one-dimensional models.

Since the models are essentially one-dimensional, we also have a linear relationship using the operators B_1 and B_2. If $f(a, t)$ is the function that goes together with $q(x, a, t)$, then the function $F(a_1, a_2, t)$ that goes together with $Q(x_1, x_2, a_1, a_2, t)$ is given by

$$F(a_1, a_2, t) = f(a_2, a_1 t)\,. \tag{3.75}$$

Then we have $B_2(FQ) = 0$ and consequently, $B_1 B_2(FQ) = 0$. We wrap this section up with the following proposition.

Proposition 3.33. *All 64 reducible models are equivalent to ones confined to an axis. Their generating functions Q and their functions F, such that $B(FQ) = 0$, are given by one-dimensional ones with t scaled by one of a_1 or a_2.*

3.4.2 Kernel Group Considerations

We can partition the two-dimensional lattice walk models in another way — with respect to the nature of the group operations in G_S. We will combine cases for analysis in the form "case $i.j$": we have case i for x_1 and case j for x_2, from the section on groups. Of course, case $i.j$ is equivalent to case $j.i$. From the analysis in the previous sections, cases 1.1, 1.2 and 1.3 only feature reducible models, so we may skip them. We are left with the following cases

Case 2.2: In this case, both x_1 and x_2 can only be transformed via the identity transformation. Therefore, exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. The models where $S_1^1 = 0$ or $S_2^1 = 0$ already have solutions since they are reducible. The only remaining unsolved case is when $S_1^{-1} = 0$ and $S_2^{-1} = 0$. The possible models are the ones with steps in {N, E, NE} except the ones going either only in the N direction or only in the E direction. Here are the models in the positive quadrant which we have not solved yet.

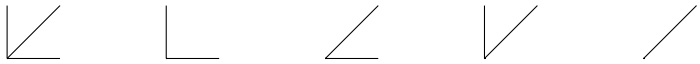


Figure 3.3: Models from case 2.2.

Case 2.3: In this case, there is only one transformation for x_1 and two for x_2. Therefore, $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case) but not both. The negative case is reducible, so we only seek solutions for the positive one. Furthermore, $S_2^{-1} \neq 0$ and $S_2^1 \neq 0$, so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. If both NE and SE are in $\mathcal{S}$, then we have 8 cases. If one of NE and SE is in $\mathcal{S}$ but not both, then we have another 8 cases. Finally, if neither is in $\mathcal{S}$, we must have the other three steps present in order to satisfy the conditions. We have 17 cases in total, the group size is two, and none of the models are reducible. Here are all of these models.

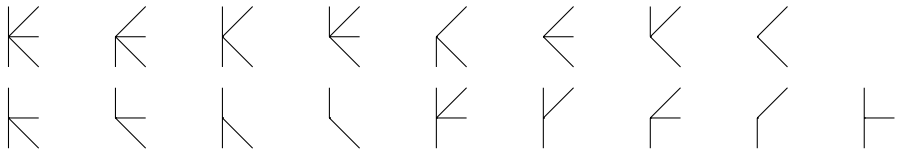


Figure 3.4: Models from case 2.3.

Case 3.3: This is the most interesting case, with two transformations for x_1 and x_2 each. Thus, there must be at least one step going in a positive and one going in a negative direction, for each of x_1 and x_2. If we look at which diagonal steps are present, similarly to case 2.2, we find that there are 145 such models. Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity, depending on the step set in question. None of these cases are reducible.

3.5 Solving Basic Two-dimensional Models

3.5.1 Case 2.2 – Group of Order One

The only models that are not reducible are those where $S_1^{-1} = 0$ and $S_2^{-1} = 0$. Here

$$S = \delta + \alpha x_1 + \beta x_2 + \gamma x_1 x_2 \tag{3.76}$$

for some α, β, γ and δ. The functional equation and its zeroth term extractions are given by

$$\begin{aligned}(1 - tS)Q - c_1 Q_1^0 - c_2 Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} &= \overline{a}_1 \overline{a}_2 \,, \\ (1 - ta_1(\delta + \beta x_2))Q_1^0 - c_2 Q_{1,2}^{0,0} &= \overline{a}_2 \,, \\ (1 - ta_2(\delta + \alpha x_1))Q_2^0 - c_1 Q_{1,2}^{0,0} &= \overline{a}_1 \,, \\ (1 - a_1 a_2 t\delta)Q_{1,2}^{0,0} &= 1 \,.\end{aligned} \tag{3.77}$$

This is a linear system for $Q_{1,2}^{0,0}$, Q_1^0, Q_2^0 and Q, and by back-substitution,

$$Q = \frac{1}{1 - tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1 - ta_1a_2\delta}\left(c_1\frac{1 - a_1\delta t}{1 - ta_1 S_1^0} + c_2\frac{1 - a_2\delta t}{1 - ta_2 S_2^0} - c_1c_2\right)\right) \tag{3.78}$$

$$= \frac{1}{1 - tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1 - a_1a_2\delta t}(c_1 f_1(a_1, t) + c_2 f_2(a_2, t) - c_1c_2)\right) \,. \tag{3.79}$$

Note how Q depends on functions f_i that are each missing one of the a variables. We shall see that the linear relation $B(FQ) = 0$ is satisfied by the function

$$F(a_1, a_2, t) = 1 - a_1a_2\delta t \,. \tag{3.80}$$

First,

$$B((1 - a_1a_2\delta t)\overline{a}_1\overline{a}_2) = \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1c_2 \,, \tag{3.81}$$

$$B(c_1f_1 + c_2f_2 - c_1c_2) = c_1f_1 + c_2f_2 - c_1c_2 - c_2f_2 - c_1f_1 = -c_2c_2 \,. \tag{3.82}$$

Putting the two quantities together, along with dividing by $1 - tS$, shows that $B(FQ) = 0$. Of course, this works for $Q_{1,2}^{0,0}$ too.

3.5.2 Case 2.3 – Group of Order Two

In this case only S_1^{-1} is zero, so the functional equation are

$$\begin{aligned}(1-tS)Q - c_1Q_1^0 - (c_2 - t\overline{x}_2S_1^{-1})Q_2^0 + c_1c_2Q_{1,2}^{0,0} &= \overline{a}_1\overline{a}_2\,,\\ (1-ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x}_2S_{1,2}^{0,-1})Q_{1,2}^{0,0} &= \overline{a}_2\,,\\ (1-ta_2S_2^0)Q_2^0 - c_1Q_{1,2}^{0,0} - a_2tS_2^{-1}Q_2^1 &= \overline{a}_1\,,\\ (1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2S_{1,2}^{0,-1}Q_{1,2}^{0,1} &= 1\,.\end{aligned} \tag{3.83}$$

We can identify a subproblem that is essentially one-dimensional and therefore, already solved. To see this, take the second and fourth equations, and make the six substitutions $a_i = a$, $c_i = c$, $x_2 = x$, $a_1t = \tau$, $S_1^0 = T$ and $Q_1^0(x_2) = q(x)$. The two equations become

$$(1-\tau T)q - (c - \tau\overline{x}T_{-1})q_0 = \overline{a}\,, \tag{3.84}$$

$$(1-a\tau T_0)q_0 - a\tau T_{-1}q_1 = 1\,. \tag{3.85}$$

We can see that this is a one-dimensional problem again. The solution here is $q(x, a; \tau)$, obtained by the methods in the previous chapter, so the original problem has solutions

$$Q_1^0(x_2) = q(x_2, a_2; a_1t) \quad \text{and} \quad Q_{1,2}^{0,0} = q_0(a_2; a_1t)\,. \tag{3.86}$$

Since $Q_{1,2}^{0,0}$ and Q_1^0 are given by a one-dimensional model's solution, they have a linear relationship with their partially-marked versions. Next, we rewrite the master equation as

$$(1-tS)Q - (c_2 - t\overline{x}_2S_2^{-1})Q_2^0 = \overline{a}_1(\overline{a}_2 + b_1q - b_1c_2q_0) = \overline{a}_1\phi(x_2)\,. \tag{3.87}$$

Since $S_1^{-1} \neq 0$ and $S_2^1 \neq 0$, we have a root $x_2 = \sigma_2$ that allows us to eliminate the $(1-tS)Q$ term. That is, we are finding a root for x_2 again, since $x = x_2$ in the previous equations. We find that

$$Q_2^0 = -\frac{\overline{a}_1\phi(\sigma_2)}{c_2 - t\overline{\sigma}_2S_2^{-1}} \tag{3.88}$$

and

$$Q = \frac{1}{1-tS}\left(\overline{a}_1\phi(x_2) - \frac{c_2 - t\overline{x}_2S_2^{-1}}{c_2 - t\overline{\sigma}_2S_2^{-1}}\overline{a}_1\phi(\sigma_2)\right)\,. \tag{3.89}$$

We may simplify this form based on what q and q_0 are from the one-dimensional setting. We will do something else. After a quick computation, we have $Q_2^0 = \overline{a}_1 a_2 \phi(\sigma_2) \widehat{Q}_2^0$ and we may write

$$KQ = \overline{a}_1 \phi + K_2 \overline{a}_1 a_2 \phi(\sigma_2) \widehat{Q}_2^0 \,, \tag{3.90}$$

$$K\widehat{Q} = \overline{a}_2 + K_2 \widehat{Q}_2^0 \,. \tag{3.91}$$

Therefore, after expanding the definitions of the various quantities, we have

$$KQ - \overline{a}_1 a_2 \phi(\sigma_2) K\widehat{Q} = \overline{a}_1(\phi - \phi(\sigma_2)) = c_1(q - q(\sigma_2)) \tag{3.92}$$

Since $q = q(x_2, a_2; a_1 t)$, there is a function f depending on a's and t only, for which $B_2(fq) = 0$. The same function works for $q(\sigma_2)$ as well. Finally, there is a linear relationship between Q and its various partially-interacting versions. This time it seems that the relationship is not given by $B(FQ) = 0$ but rather a more complicated one, $B_2(fg(B_1(Q/g))) = 0$, where $g = \overline{a}_1 a_2 \phi(\sigma_2)$. Finally, we may give the following theorem.

Theorem 3.34. *Consider the two-dimensional classic models. If the model is reducible or has a group of order one or two, then the generating function depends on its partially-interacting versions in a linear manner.*

Miscellaneous Case

There is one case with an infinite group that is both easy to solve and not reducible — the model with $S = x_1 x_2 + \overline{x}_1 \overline{x}_2$. The transformations are $g_1 = \overline{x}_1 \overline{x}_2^2$ and $g_2 = \overline{x}_2 \overline{x}_1^2$. If we apply them on (x_1, x_2) in an alternating sequence, each time the absolute powers of x_1 and x_2 will increase in one of the coordinates, and therefore, the group is infinite. However, this case is easy to solve because the model is essentially one-dimensional. It is equivalent to the model on the non-negative line with $S = x + \overline{x}$ and marking visits to the origin with a, where $x = x_1 x_2$ and $a = a_1 a_2$. The solution is

$$Q(x_1, x_2, a_1, a_2; t) = q(x_1 x_2, a_1 a_2; t) \tag{3.93}$$

and we can add this model to the theorem as well.

3.5.3 Counting Classes of Models

We will tally all the models on the quarter plane. We have 64 reducible models, 5 new cases with group of order one and 34 new cases with group of order two. Also, there are 29 models with group of order four, 6 with order six, and 4 with order eight, plus one easy case with infinite group. So

far we have 143 cases of models we can solve easily or models with a finite group, some of which we can solve easily. Next, by inspection of Table 4 in [8, p. 28], we see there are 120 models with infinite groups: 12 with three steps, 40 with four steps, 44 with five steps, and 24 with six steps. Adding all of those together yields 263 cases. There are 7 reducible models with an infinite group that we have counted twice. The total count is 256 and we are done.

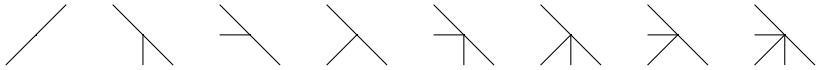


Figure 3.5: Reducible models with an infinite group.

So far we have solved the 64 reducible models, the 39 other models with group of order one or two, and the single new case of easy-to-solve model with an infinite group, for a grand total of 104 trivial or easy models. Out of the remaining 152 models, 113 have an infinite group, and their analysis will not be a part of this thesis. Finally, we have 39 cases left, with a group of order four, six or eight. The rest of the thesis will handle the analysis of those cases.

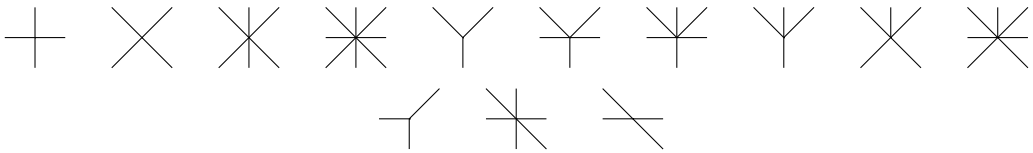


Figure 3.6: Models with a finite group.

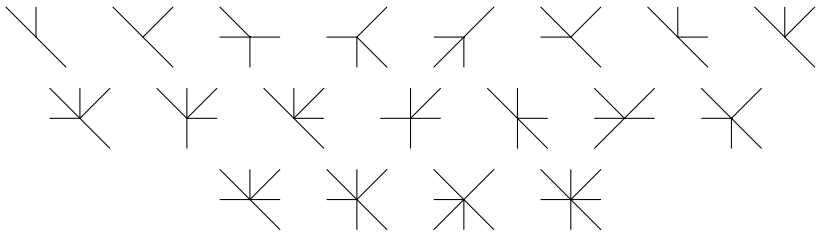


Figure 3.7: Models with an infinite group. To get the rest of them, apply one of the symmetries of the square.

Chapter 4

The Algebraic Kernel Method

In this chapter we will explore the algebraic kernel method as seen in Section 4 of [18, p. 9] and we will extend it to handle models with interactions. First, we will recast the non-interacting version of the method in a form that uses operators. Then we will generalize the operators by giving them weights, and this will allow us to investigate models with interactions.

4.1 Models Without Interactions

At the core of the algebraic kernel method we have the idea that we can apply transformations from a group G onto the master functional equation and acquire different equations for each $g \in G$. After we sum these equations in a nice way, we manage to eliminate the functions Q_1^0, Q_2^0, $Q_{1,2}^{0,0}$ and their orbits, only leaving the orbit of the kernel term KQ and a term on the right-hand side. What makes this possible is pre-multiplying the whole equation by x_1x_2. The analysis also works in higher dimensions, as we will show later. Although the generalization has been implemented before (for example,[8, p. 11] for the two-dimensional case and [18, p. 10] for the three-dimensional one), we will present it in a compact and easy-to-generalize way. Instead of summing equations as in [8], we will use an operator D_G that will do the same thing. We will define D_G in a way that is analogous to the way the algebraic kernel method is currently applied. We will focus on groups with a full set of d generators.

Definition 4.1. *For a lattice walk model with small steps, let G be the group generated by the transformations $g_k : x_k \mapsto \overline{x}_k S_k^{-1}/S_k^1$ preserving the kernel, and for g in G, let $|g|$ denote the sign of g. Finally, let D_G be defined as*

$$D_G = \sum_{g \in G} (-1)^{|g|} g \,. \tag{4.1}$$

In two dimensions we can find the whole group by iterating over the two generators. That is, we apply g_1 and g_2 sequentially and the group is $\{1,\, g_i,\, g_j g_i,\, g_i g_j g_i, \ldots\}$. However, we shall use the equivalent orbit $\{1,\, g_i,\, g_i g_j,\, g_i g_j g_i, \ldots\}$, where the iteration is on the right. To declutter the notation, we will use g_{iji} instead of $g_i g_j g_i$.

Definition 4.2. *For a sequence $I = (i_1, \ldots, i_n)$ of positive integers, define*

$$g_L = g_{i_1,\ldots,i_n} = g_{i_1} \cdots g_{i_n}\,. \tag{4.2}$$

If the group is finite for a two-dimensional classic model, then $(g_1 g_2)^n = 1$ for some minimal n, so we can rewrite D_G in a useful way. The following lemma is a restatement of the orbit summation method, found in [8, p. 10], recast in operator notation.

Lemma 4.3. *For a two-dimensional classic model with group of order $2n$, generated by g_1 and g_2, we can write the operator D_G as*

$$D_G = (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i)\,. \tag{4.3}$$

Proof. We can just check the identity by expanding it as

$$\begin{aligned} D_G &= (1 + g_{ij} + \cdots + (g_{ij})^{n-1})(1 - g_i) \\ &= 1 - g_i + g_{ij} - g_{iji} + \cdots - (g_{ij})^{n-1} g_i\,, \end{aligned} \tag{4.4}$$

giving us the orbit summation resulting from the second form of the obit. □

Here are some examples of small groups.

Example 4.4. When the group has order four, the elements commute and the orbit is given by $\{1,\, g_1,\, g_{12} = g_{21},\, g_2\}$. The operator is $D_G = (1 + g_{12})(1 - g_1)$, but it also factors in a convenient way as

$$D_G = 1 - g_1 - g_2 + g_{12} = (1 - g_1)(1 - g_2)\,. \tag{4.5}$$

When the group has order six, the orbit is $\{1,\, g_1,\, g_{21},\, g_{121} = g_{212},\, g_{12},\, g_2\}$. We can write the operator as

$$D_G = (1 + g_{12} + g_{1212})(1 - g_1) = 1 - g_1 - g_2 + g_{12} + g_{21} - g_{121}\,. \tag{4.6}$$

We can generalize this result for the groups of higher-dimensional models.

Lemma 4.5. *Consider a d-dimensional classic model with group G. The set H of elements with even parity forms a subgroup.*

Proof. The proof is analogous to the proof that the set of even permutations forms a subgroup of the set of permutations of $\{1, \ldots, n\}$. First, the identity i of G is in H, because $i^2 = i$ has even parity. Then H is closed under the group operation because the product of two elements with even parity also has even parity. Finally, if x has even parity, then its inverse y has even parity, otherwise, $i = xy$ would not have even parity. Thus, H is a subgroup by the two-step subgroup test. □

The next corollary should be self-evident.

Corollary 4.6. *Let g_i and g_j be generators of G. Then $g_{ij}(H)$ and $g_i(H) = g_j(H)$.*

Lemma 4.7. *Consider a d-dimensional classic model with finite group G. For a nontrivial generator g_k, we have $D_G = D_H(1 - g_k)$, where H is the subgroup of even elements.*

Proof. First, note that g_k is an involution that does not fix any element of G, that is $gg_k \neq g$ for any g in G. Since applying g_k on a group element changes its parity, this means that g_k induces a partition of G into two parts: a set of even elements and a set of odd elements in one-to-one correspondence. That is, we can partition G as the union of H and Hg_k. With the notation $e_h = (-1)^{|g|}$, we can immediately compute the form of D_G from the statement of the lemma:

$$D_G = \sum_{g \in H} e_g g + \sum_{g \in H} (-1) e_g g g_k = \left(\sum_{g \in H} e_g g \right) (1 - g_k) = D_H (1 - g_k) \,. \tag{4.7}$$

□

Note that we could have also computed $D_G = (1 - g_k) D_H$ if we multiplied g_k on the left. With the previous generalization of the operator form, we obtain the d-dimensional generalization to Proposition 5 in [8, p. 11] and Lemma 52 in [15, p. 59].

Theorem 4.8. *For a d-dimensional classic model with a finite group G,*

$$D_G(x_1 \cdots x_d K Q) = D_G(x_1 \cdots x_d) \,. \tag{4.8}$$

Proof. With the application of $D_G x_1 \cdots x_d$, all terms except KQ and 1 vanish since terms like $x_{[d]} \overline{x}_I t S_I^{-\mathbb{1}} Q_I^{\mathbb{0}}$ are annihilated by D_G because they miss one or more variable x_i. □

In [8, p. 11] and [18, p. 13], the authors have found that sometimes the algebraic kernel method fails. That is, $D_G(x_1 \cdots x_d) = 0$ and the equation does not give us new information. The operator form of the method allows us to find a more general instance of when this situation occurs.

Corollary 4.9. *If an element h of G is an involution that permutes the variables $x_1, \ldots, x_d$, then the regular algebraic kernel method fails. That is,*

$$D_G(x_1 \cdots x_d KQ) = 0\,. \tag{4.9}$$

Proof. Since h is an involution, the same method applies as before and we can factor the operator D_G as $D_H(1-h)$ for some group H in G. The action $1-h$ eliminates the monomial $x_1 \cdots x_d$. □

Example 4.10. In two dimensions, the algebraic kernel method fails precisely for those models where we have a permutation as in the corollary. That is, $h(x_1, x_2) = h(x_2, x_1)$ and h has order two. Proposition 6 in [8, p. 12] follows.

For models with six steps in three dimensions, the algebraic kernel method fails in 62 of them [18, p. 13]. Here G contains a permutation $h(x_i, x_j) = (x_j, x_i)$. It is not clear if all cases of failure of the method can be attributed to such a situation.

Naturally, we need to find out if we can use D_H instead of $D_G = D_H(1-h)$? The answer is yes. The operator D_H eliminates some of the Q variables and introduces others, but the result is a functional equation that is symmetric under permutation by h. One may hope to perform a method analogous to the half-orbit summation method to prove algebraicity or D-finiteness.

Next, we will generalize the orbit summation operator by including weights with each g in G.

Definition 4.11. *Given a group G acting on $x_1, \ldots, x_d$ and weights R_g for each g in G, the weighted orbit summation (or orbital) operator is defined as*

$$D = \sum_{g \in G} e_g R_g g\,. \tag{4.10}$$

The normal orbit summation operator is one with weights $R_g = 1$. The action of such an operator can be seen in [21, p. 11], where the authors use an orbit summation with weights. We can capture various possible strategies in this notation.

Lemma 4.12. *Pre-multiplying the functional equation by a function T is equivalent to applying a weighted orbital operator on the original equation. Application of more than one weighted operator is equivalent to applying a single weighted operator.*

Proof. For the first statement, let T and f be some functions. Then we have

$$D_G(Tf) = \sum_{g \in G} e_g g(Tf) = \sum_{g \in G} e_g g(T) g(f) = \sum_{g \in G} e_g R_g g(f)\,, \tag{4.11}$$

where the weights are $R_g = g(T)$. For the second statement, we multiply two weighted operators as

$$D_1 D_2 = \sum_{g\in G}\sum_{g\in G} e_g R_g g e_h T_h h = \sum_{g\in G}\sum_{h\in G} R_g e_{gh} T_{gh} g h = \sum_{g\in G}\left(\sum_{h\in G} e_{gh^{-1}} R_{gh^{-1}} T_g\right) g \qquad (4.12)$$

and we obtain a new weighted operator. □

The second statement of the lemma leads to the following proposition.

Proposition 4.13. *The set of weighted orbit summation operators, associated with a group G, forms a monoid under composition.*

Proof. Composition of operators is associative and in the last example we showed that the set is closed under composition. Thus we only need to show that the set has an identity. Let $R_g = 0$ for all g in G except for $R_1 = 1$. Then D_G for these weights is the identity operator. □

Because of the preceding lemma, if we can circumvent the failure of the algebraic kernel method via a clever choice of function T or two operators D_1 and D_2, then we should focus on using a weighted operator from the start. The procedure of eliminating terms from the functional equation requires that those terms be removed generically. This means that whenever $I \neq J$, the functions $Q_I^{\emptyset}$ and $Q_J^{\emptyset}$ are treated as independent variables in a linear combination and thus, each of them needs to be eliminated by the operator on its own. That is, the procedure requires $D'(t\overline{x}_I S_I^{-1} Q_I^{\emptyset}) = 0$ for each possible nonempty set I. This allows us to show that in general, when the algebraic kernel method fails, all the operators D' that eliminate the unwanted terms in the functional equation also eliminate the monomial $x_1 \cdots x_d$. It is enough to work on the functional equation $x_1 \cdots x_d KQ = x_1 \cdots x_d$ with some operator D. Then we can find the original operator D' from the relation $D' = Dx_1 \cdots x_d$, where D' acts on the master functional equation by pre-multiplying it by $x_1 \cdots x_d$ and applying D. We will show that all the weights of D must be the same.

This observation turns the task of finding the weights into a linear algebra problem. We just need to apply D on the functions $W_I = t\overline{x}_I S_I^{-1} Q_I^{\emptyset}$, which are missing the variables indexed by I, collect the different results and set their coefficients equal to zero. That is, for every $f = W_I$ we find the orbit

$$\underset{G}{\mathrm{Orbit}}(f) = \{g(f) | g \in G\} \qquad (4.13)$$

and then find

$$0 = D(f) = \sum_{g\in G} R_g g(f) = \sum_{f'\in Orbit(f)} R'_g f' \,, \qquad (4.14)$$

where we set

$$R'_g = \sum_{\substack{g(f)=f' \\ g\in G}} R_g = 0\,. \tag{4.15}$$

Thus, we reach a linear system of $|\operatorname{Orbit}_G(f)|$ equations in $|G|$ unknowns. We can do this for each of the $2^d - 1$ terms W_I and solve the system of equation. Here is an example

Example 4.14. Consider the two-dimensional classic model with $S = \overline{x}_1 + \overline{x}_2 + x_1x_2$, where the algebraic kernel method fails [8, p.12]. Here both of the generators map x_i to $\overline{x}_1\overline{x}_2$ and the group is given by the orbit $\{1, g_1, g_{21}, g_{121}, g_{12}, g_2\}$ of (x_1, x_2). The orbit of 1 under G is trivial and by symmetry of the group actions, the orbit of each of $W_1 = f_1(x_2)$ and $W_2 = f_2(x_1)$ is given by $\{f(x_1),\, f(x_2),\, f(\overline{x}_1\overline{x}_2)\}$. The relations we obtain are

$$R - R_1 + R_{21} - R_{121} + R_{12} - R_2 = 0\,, \tag{4.16}$$

$$(R - R_2)f(x_1) + (R_{21} - R_{121})f(x_2) + (R_{12} - R_1)f(\overline{x}_1\overline{x}_2) = 0\,, \tag{4.17}$$

$$(R_{12} - R_{121})f(x_1) + (R - R_1)f(x_2) + (R_{21} - R_2)f(\overline{x}_1\overline{x}_2) = 0\,, \tag{4.18}$$

and if we set the coefficients to zero, we find that all the weights are the same.

Lemma 4.15. *The stabilizer of x_k is given by the subgroup generated by g_i for $i \neq k$.*

Proof. We already know that g_i fixes x_k for every $i \neq k$. Consider an element $g = g_{a_1\ldots a_n}$ in G that fixes x_k. For a a given g we may obtain a shorter action g' by the following procedure. If a_1 or a_n is not k, then we may take $g_{a_1}gx_k = g_{a_2\ldots a_n}x_k = g_{a_1}x_k = x_k$. Also, if both a_1 and a_n are equal to k, we may take $g_kgg_kx_k = g_kx_kg_k = x_k$. If we follow this procedure to its conclusion we either obtain 1 or g_k. However, the latter one does not fix x_k so we must arrive at $g = 1$ with this procedure, and we have just proved that every g that fixes x_k must have an even number of g_k's in its generating sequence. Somehow, it turns out that such actions are equivalent to ones with no g_k's in their representation. That is, if g fixes x_k it has no g_k's in its generating set, so we are done. □

Lemma 4.16. *The stabilizer of W_k is given by $\{1, g_k\}$.*

Proof. If g stabilizes the generic function W_k, then it must also stabilize each x_i for $i \neq k$. Then g must be in the stabilizer of x_i for every $i \neq k$. The intersection of these sets is precisely $\{1, g_k\}$. □

Corollary 4.17. *The actions h in H lead to distinct functions $h(W_k)$.*

Proof. First, because $h(W_k) = hg_k(W_k$ for every h in H, the orbit of W_k^0 is the same whether we use H or G. By the orbit stabilizer theorem, the orbit of W_k^0 under G has size $|G|/2 = |H|$, so all the elements $h(W_k)$ must be distinct for different h's. □

Lemma 4.18. *Consider a weighted orbital operator that annihilates functions W_k for all k. Then its weights satisfy the equations $R_h = R_{hg_k}$ for all h in H.*

Proof. We just compute the action of the orbital operator as

$$D(W_k^0) = \sum_{g \in G} (-1)^{|g|} R_g g(W_k) = \sum_{h \in H} (R_h - R_{hg_k}) h(W_k) \tag{4.19}$$

and since the functions $h(W_k)$ are distinct, their coefficients must be identically zero as well. □

Theorem 4.19. *Consider a weighted orbital operator that annihilates functions W_k for all k. Then all of its weights are the same.*

Proof. From the lemma, we see that $R_h = R_{hg_k}$ for k and all h in H. That is, the weights are the same if their corresponding group elements differ by a generator g_k. Thus, since we can reach any g in G by applying a sequence of generators, $R_g = R_1$ for all g, and the weights are all the same. □

This has the implication that there is essentially one weighted orbital operator that annihilates all the functions W_k^0 and therefore, there is no clever weighted operator that would allow us to circumvent the failure of the algebraic kernel method for classic models.

Corollary 4.20. *We cannot circumvent the failure of the algebraic kernel method through the use of a weighted orbital operator.*

4.2 Models With Interactions

It is natural to consider extending the algebraic kernel method to one working for interacting models, but we quickly see that the regular orbit summation operator is not sufficient. This is where the power of the weighted orbital summation operators starts to show. The analysis for two-dimensional models is particularly nice because we end up having the same number of equations as unknowns, and we are able to find the required weights. The linear systems can be written as matrix equations, where the determinant of the matrix tells us when we have a useful orbit summation operator. In the end, the method does not extend fully because we do not eliminate the $Q_{1,2}^{0,0}$ term. The D-finiteness of the full generating function Q depends on the D-finiteness of the function $Q_{1,2}^{0,0}$. We will only consider the models with finite groups that have two nontrivial generators. Therefore, the group G has order 4, 6 or 8, and the functional equation in two dimensions is

$$KQ - K_1 Q_1^0 - K_2 Q_2^0 + K_{1,2} Q_{1,2}^{0,0} = \alpha\,. \tag{4.20}$$

4.2.1 Groups of Order Four

The orbital summation operator for the non-interacting case is

$$D_G = 1 - g_1 - g_2 + g_1 g_2 = (1 - g_1)(1 - g_2) = \Delta_1 \Delta_2 \tag{4.21}$$

but this operator does not eliminate the unknown functions in the interacting case. Instead of using this operator, we will use a weighted one and we will determine its weights. We will find a linear system of equations by applying the group operations on the original functional equation. The unknowns will be functions Q_1^0, Q_2^0 and their orbits under G. If the associated matrix has a nonzero determinant, the system has a unique solution and unfortunately, we will not be able to eliminate variables. However, there are some cases where the determinant is actually zero, so we can eliminate some of the unknowns.

Since the transformations commute, we have $g_i Q_i^0 = Q_i^0$ and $g_i g_j Q_i^0 = g_j Q_i^0$. The variables and their orbits are given by the next table.

Group action	Effect on Q_1^0	Effect on Q_2^0
1	Q_1^0	Q_2^0
g_1	Q_1^0	$g_1 Q_2^0$
g_2	$g_2 Q_1^0$	Q_2^0
$g_{12} = g_{21}$	$g_2 Q_1^0$	$g_1 Q_2^0$

Table 4.1: Effects of group actions $g \in G$ on Q_1^0 and Q_2^0.

Therefore, the only variables that we need to eliminate are in the set

$$E = \{Q_1^0,\, Q_2^0,\, g_2 Q_1^0,\, g_1 Q_2^0\}\,. \tag{4.22}$$

The situation looks promising since we also have four equations. We rewrite the functional equation in a way that highlights those variables.

$$K_1 Q_1^0 + K_2 Q_2^0 = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha = \beta\,. \tag{4.23}$$

Then we apply the transformations to produce the system of equations

$$K_1 Q_1^0 + K_2 Q_2^0 = \beta\,, \tag{4.24}$$

$$g_1 K_1 Q_1^0 + g_1 K_2 g_1 Q_2^0 = g_1\beta = \beta_1\,, \tag{4.25}$$

$$g_2 K_1 g_2 Q_2^0 + g_2 K_2 Q_2^0 = g_2\beta = \beta_2\,, \tag{4.26}$$

$$g_{12} K_1 g_2 Q_1^0 + g_{12} K_2 g_1 Q_2^0 = g_{12}\beta = \beta_{12}\,. \tag{4.27}$$

Using the order of variables in E, we can write the associated matrix M of the linear system as

$$M_4 = \begin{bmatrix} K_1 & K_2 & 0 & 0 \\ g_1 K_1 & 0 & 0 & g_1 K_2 \\ 0 & g_2 K_2 & g_2 K_1 & 0 \\ 0 & 0 & g_{12} K_1 & g_{12} K_2 \end{bmatrix}. \tag{4.28}$$

Then

$$\begin{aligned} \det M_4 &= K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 K_1 g_2 K_1 K_2 g_{12} K_2 \\ &= K_1 g_{12} K_1 g_1 K_2 g_2 K_2 - g_1 (g_1 K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \\ &= \Delta_1 (K_1 g_{12} K_1 g_1 K_2 g_2 K_2) \end{aligned} \tag{4.29}$$

where the last equation is obtained by using g_2 instead of g_1 in the factoring.

The determinant must vanish if we are to eliminate variables, and that only happens when the expression in Δ_1 is in the operator's kernel. The models with groups of order four exhibit at least one symmetry across a line — either vertical or horizontal symmetry, or both. Without loss of generality, let the model have symmetry across the x_2 axis. That is, $g_1(x_1) = x_1$, $g_1(S_2^i) = S_2^i$ and $g_1(S_1^i) = S_1^{-i}$, and we can compute some of the quantities. For example, $g_1 K_2 = K_2$ and $g_1 g_2 K_2 = g_2 g_1 K_2$. Therefore, Δ_1 fixes both K_2 and $g_2 K_2$ and

$$\det M_4 = \Delta_1 (K_1 K_2 g_{12} K_1 g_2 K_2) = K_2 g_2 K_2 \Delta_1 (K_1 g_{12} K_1)\,. \tag{4.30}$$

This quantity is zero if and only if $\Delta_1(K_1 g_{12} K_1) = 0$, so we have

$$\begin{aligned} \Delta_1 (K_1 g_{12} K_1) &= K_1 g_{12} K_1 - g_1 K_1 g_2 K_2 \\ &= c_1 t (\overline{x}_1 - x_1)(g_2 S_1^{-1} - S_1^{-1}) \\ &= -c_1 t (\overline{x}_1 - x_1) \Delta_2 (S_1^{-1})\,. \end{aligned} \tag{4.31}$$

Therefore, this equation is true if and only if $g_2 S_1^{-1} = S_1^{-1}$. We have

$$g_2(S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}x_2) = S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,0} + S_{1,2}^{-1,1}\overline{x}_2\mu_2 \tag{4.32}$$

$$S_{1,2}^{-1,-1}x_2\overline{\mu}_2 + S_{1,2}^{-1,1}\overline{x}_2\mu_2 = S_{1,2}^{-1,-1}\overline{x}_2 + S_{1,2}^{-1,1}x_2\,, \tag{4.33}$$

and we can write it succinctly as $S_{1,2}^{-1,-1} = \mu_2 S_{1,2}^{-1,1}$.

If one of $S_{1,2}^{-1,-1}$ and $S_{1,2}^{-1,1}$ is zero, then so is the other, and since the step set has vertical symmetry, $S_{1,2}^{1,1}$ and $S_{1,2}^{1,-1}$ are also zero. That is, the model has no diagonal steps. However, since the group has two nontrivial generators, we must have both steps with monomials $\overline{x}_2$ and x_2, and the model is highly-symmetric. If both of the S values are equal to one, then $S_2^{-1} = S_2^1$ and again, the model is highly-symmetric.

Conversely, if we started with a highly-symmetric model, we would have obtained $\det M_4 = 0$, and this leads to the following theorem.

Theorem 4.21. *The determinant* $\det M_4 = 0$ *if and only if the model is highly-symmetric.*

Corollary 4.22. *The algebraic kernel method extends to the non-interacting case for* $|G| = 4$ *if and only if the model is highly-symmetric.*

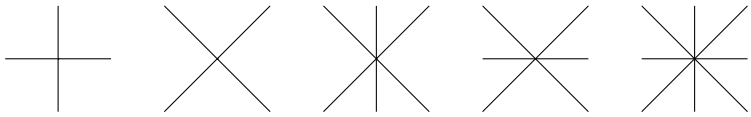


Figure 4.1: The highly-symmetric models on the plane with two nontrivial generators for the group G.

We are finally ready to use the orbit summation method. There are 8 highly-symmetric two-dimensional models, but only five of them have two nontrivial generators for the group G. Furthermore, two of the models are reflections of each other, so we are really left with four distinct cases. From left to right in the figure above, they are the Cartesian model, the Diagonal Cartesian model, the model with two horizontal steps missing, and the full model. If we are able to eliminate all the variables from the set E in the beginning of this section, then we will have a linear relation like

$$\gamma\beta - \gamma_1\beta_1 - \gamma_2\beta_2 + \gamma_{12}\beta_{12} = 0\,. \tag{4.34}$$

Here the choice of signs is inspired by the orbit summation operators. Let $\bullet$ and $*$ stand for applications of g_1 and g_2, respectively. Note that we have $K_2^\bullet = K_2$ and $K_1^* = K_1$. Next, we let

$$x = Q_1^0\,, \quad y = Q_2^0\,, \quad u = Q_1^{0,*}\,, \quad v = Q_2^{0,\bullet}\,, \tag{4.35}$$

$$a = K_1\,, \quad b = K_1^\bullet\,, \quad c = K_2\,, \quad d = K_2^*\,. \tag{4.36}$$

The original equations attain a simplified format given by

$$\beta = ax + cy\,, \quad \beta_1 = bx + cv\,, \tag{4.37}$$

$$\beta_2 = dy + au\,, \quad \beta_{12} = bu + dv\,, \tag{4.38}$$

Pluggin these into the linear relation above yields

$$\begin{aligned} 0 &= (ax + cy)\gamma - (bx + cv)\gamma_1 - (dy + au)\gamma_2 + (bu + dv)\gamma_{12} \\ &= (a\gamma - b\gamma_1)x + (c\gamma - d\gamma_2)y + (b\gamma_{12} - a\gamma_2)u + (d\gamma_{12} - c\gamma_1)v\,, \end{aligned} \tag{4.39}$$

and in order to eliminate the variables x, y, u and v, we need all their coefficients to vanish. We obtain the equations

$$a\gamma = b\gamma_1\,, \quad c\gamma = d\gamma_2\,, \quad b\gamma_{12} = a\gamma_2,\,, \quad d\gamma_{12} = c\gamma_1\,. \tag{4.40}$$

If we set $\gamma = 1$, then $\gamma_1 = a/b$ and $\gamma_2 = c/d$ and $\gamma_{12} = (ac)/(bd)$. So we can multiply them out by bd to find the relation

$$bd\beta - ad\beta_1 - cb\beta_2 + ac\beta_{12} = 0\,, \tag{4.41}$$

where we have successfully eliminated Q_1^0, Q_2^0, $Q_1^{0,*}$ and $Q_2^{0,\bullet}$. In terms of the K's we have

$$K_1^\bullet K_2^* \beta - K_1 K_2^* \beta^\bullet - K_1^\bullet K_2 \beta^* + K_1 K_2 \beta^{\bullet *} = 0\,. \tag{4.42}$$

We can rewrite this equation in the following nicer way.

Theorem 4.23. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators, we have the functional equation $\Delta(K_1^\bullet K_2^* \beta) = 0$, where $\beta = KQ + K_{1,2} Q_{1,2}^{0,0} - \alpha$.*

This is a generalization of the result in Section 4.2 of [21, p. 16], where the authors use a weighted orbit sum to eliminate boundary terms. We can expand the functional equation to produce

a different form. Let $K_0 = K_1^\bullet K_2^*$. We find that

$$\begin{aligned}\Delta(K_0\beta) &= \Delta(K_0(KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha)) \\ &= K\Delta(K_0Q) + \Delta(K_0K_{1,2})Q_{1,2}^{0,0} - \Delta(K_0)\alpha\,,\end{aligned} \tag{4.43}$$

where the second equality follows since Δ commutes with K and functions with some variables x_i missing. All that remains is to expand the various quantities involved. Then we have

$$\Delta g_i K_i = -\Delta K_i = -(c_i - tx_iS_i^{-1} - c_i + t\overline{x}_iS_i^{-1}) = -t(\overline{x}_i - x_i)S_i^{-1}\,, \tag{4.44}$$

$$\Delta K_0 = \Delta_1(K_1^\bullet)\Delta_2(K_2^*) = t^2(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)S_1^{-1}S_2^{-1}\,. \tag{4.45}$$

Next,

$$\Delta(K_0K_{1,2}) = \Delta_1K_1^\bullet\Delta_2K_2^*K_{1,2} = \Delta_1K_1^\bullet(K_2^*K_{1,2} - K_2K_{1,2}^*)\,, \tag{4.46}$$

where, after expanding, the quantity in parentheses is

$$\begin{aligned}K_2^*K_{1,2} - K_2K_{1,2}^* =& c_1c_2tS_2^{-1}(\overline{x}_2 - x_2) - c_2t\overline{x}_2S_{1,2}^{-1,-1}(\overline{x}_2 - x_2) \\ =& c_2t(\overline{x}_2 - x_2)(c_1S_2^{-1} - \overline{x}_1S_{1,2}^{-1,-1})\,.\end{aligned} \tag{4.47}$$

Finally,

$$\Delta(K_0K_{1,2}) = c_2t(\overline{x}_2 - x_2)\Delta_1K_1^\bullet(c_1S_2^{-1} - \overline{x}_1S_{1,2}^{-1,-1})\,. \tag{4.48}$$

To simplify this quantity, note that

$$\Delta_1K_1^\bullet c_1S_2^{-1} = c_1S_2^{-1}\Delta_1K_1^\bullet = -c_1tS_2^{-1}(\overline{x}_1 - x_1)\,, \tag{4.49}$$

$$\Delta_1K_1^\bullet\overline{x}_1S_{1,2}^{-1,-1} = S_{1,2}^{-1,-1}\Delta_1K_1^\bullet\overline{x}_1 = c_1(\overline{x}_1 - x_1)S_{1,2}^{-1,-1}\,. \tag{4.50}$$

Therefore, we find that

$$\Delta(K_0K_{1,2}) = c_1c_2t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)(tS_1^{-1}S_2^{-2} - S_{1,2}^{-1,-1})\,, \tag{4.51}$$

and after we combine terms and simplify, we can write down the following corollary.

Corollary 4.24. *For a two-dimensional highly-symmetric model with a group with two nontrivial generators and $K_0 = K_1^\bullet K_2^*$, we have the functional equation*

$$K\Delta(K_0Q) = \alpha t(\overline{x}_1 - x_1)(\overline{x}_2 - x_2)\left(tS_1^{-1}S_2^{-1}(1 - b_1b_2Q_{1,2}^{0,0}) + S_{1,2}^{-1,-1}Q_{1,2}^{0,0}\right)\,. \tag{4.52}$$

4.2.2 Groups of Order Six

We will go through this case faster because a lot of the work is similar. First, we tabulate all the unknowns Q_i^0 and their orbits under the actions of G. In the following table we write a sequence L for g_L and a reduced sequence for each unknown. For example, $g_1 Q_1^0 = Q_1^0$, so we write $\varnothing$ in the corresponding entry. For another example,

$$g_{2121} Q_1^0 = g_{212} Q_1^0 = g_{121} Q_1^0 = g_{12} Q_1^0 \,, \tag{4.53}$$

since $g_{212} = g_{121}$. The corresponding entry in the table is 12.

Sequence L	Alternative	Q_1^0	Q_2^0	K_1	K_2
1	21212	$\varnothing$	1	1	1
21	1212	2	21	21	21
121	212	12	21	121	212
2121	12	12	1	12	12
12121	2	2	$\varnothing$	2	2

Table 4.2: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

After we apply the group actions, we find six different functions that need to be eliminated. Let

$$u_1 = Q_1^0 \,, \quad u_2 = g_2 Q_1^0 \,, \quad u_3 = g_{12} Q_1^0 \,, \tag{4.54}$$

$$v_1 = Q_2^0 \,, \quad v_2 = g_1 Q_2^0 \,, \quad v_3 = g_{21} Q_2^0 \,. \tag{4.55}$$

There are twelve different K_1's and K_2's that we label sequentially as $A_1, \ldots, A_6$ and $B_1, \ldots, B_6$. The matrix associated with the linear system of equations is

$$M_6 = \begin{bmatrix} A_1 & & & B_1 & & \\ A_2 & & & & B_2 & \\ & A_3 & & & & B_3 \\ & & A_4 & & & B_4 \\ & & A_5 & & B_5 & \\ & A_6 & & B_6 & & \end{bmatrix} \tag{4.56}$$

with determinant

$$\det M_6 = A_2A_4A_6B_1B_3B_5 - A_1A_3A_5B_2B_4B_6\,. \tag{4.57}$$

The even index A's are found by applying g_1 on the odd index A's, and vice versa for the B's. Then

$$\det M_6 = \Delta_1(A_2A_4A_6B_1B_3B_5) = \Delta_1(g_1(K_1)g_{121}(K_1)g_{12}(K_1)K_2g_{21}(K_2)g_{12}(K_2))\,. \tag{4.58}$$

In the following table we show the relations between A's and B's for the six individual models with one of the two groups of order 6. $S11$, $S12$ and $S13$ have the one group, and $S21$, $S22$ and $S23$, have the other.

Model	Name	A-relations	B-relations
	S11	$A_1 = A_6$, $A_2 = A_5$, $A_3 = A_4$	$B_1 = B_4$, $B_2 = B_3$, $B_5 = B_6$
	S12	$A_1 = A_4$, $A_2 = A_3$, $A_5 = A_6$	$B_1 = B_2$, $B_3 = B_6$, $B_4 = B_5$
	S13	None	None
	S21	$A_1 = A_6$, $A_2 = A_5$, $A_3 = A_4$	$B_1 = B_2$, $B_3 = B_6$, $B_4 = B_5$
	S22	$A_1 = A_4$, $A_2 = A_3$, $A_5 = A_6$	$B_1 = B_4$, $B_2 = B_3$, $B_5 = B_6$
	S23	None	None

Table 4.3: Relations for $A = K_1$ and $B = K_2$ for groups of order six.

It is easy to check that when a model has nontrivial relation, $\det M_6 = 0$. Also, using a a computer, we can check that the determinant is nonzero in the last two cases where there are no relations. For the models with zero determinant, we seek the same type of relation on β and its orbit as in the case of groups of order four. We have the following linear relation

$$\begin{aligned} 0 =& \gamma\beta - \gamma_1\beta_1 + \gamma_{21}\beta_{21} - \gamma_{121}\beta_{121} + \gamma_{12}\beta_{12} - \gamma_2\beta_2 \\ =& \gamma(A_1u_1 + B_1v_1) - \gamma_2(A_6u_2 + B_6v_1) - \gamma_1A_2u_1 + B_2v_2) \\ &+ \gamma_{12}(A_5u_3 + B_5v_2) + \gamma_{21}(A_3u_2 + B_3v_3) - \gamma_{121}(A_4u_3 + B_4v_3)\,. \end{aligned} \tag{4.59}$$

Expanding out and collecting terms produces the system of equations

$$A_1\gamma = A_2\gamma_1\,, \qquad A_3\gamma_{21} = A_6\gamma_2\,, \qquad A_5\gamma_{12} = A_4\gamma_{121}\,, \tag{4.60}$$

$$B_1\gamma = B_6\gamma_2\,, \qquad B_5\gamma_{12} = B_2\gamma_1\,, \qquad B_3\gamma_{21} = B_4\gamma_{121}\,. \tag{4.61}$$

It does not have a nontrivial symbolic solution, so we need to use the relations here. There is a rational solution in each case, where γ is a free variable. We set γ to be the least common multiple of the denominators. We find that $K_0 = \gamma$ in each of the four cases with relations, as illustrated in the following table.

Model	S-polynomial	Weight $\gamma = K_0$	K-form
	$x_1\overline{x}_2 + \overline{x}_1 + x_2$	$A_3A_5B_6$	$g_{21}(K_1)g_{12}(K_1)g_2(K_2)$
	$\overline{x}_1x_2 + x_1 + \overline{x}_2$	$A_2A_5B_6$	$g_1(K_1)g_{12}(K_1)g_2(K_2)$
	$x_1x_2 + \overline{x}_1 + \overline{x}_2$	$A_2A_3B_5B_6$	$g_1(K_1)g_{21}(K_1)g_{12}(K_2)g_2(K_2)$
	$\overline{x}_1\overline{x}_2 + x_1 + x_2$	A_2B_6	$g_1(K_1)g_2(K_2)$

Table 4.4: Weight K_0 for the Orbit Summation operators.

Here we have picked the representations of A's and B's with the least number of transformations to reach the resulting expression. We also tried finding a possible K_0 by brute force for $S13$ and $S23$, but we did not find any among products of distinct A's and B's. It is likely that such a weight does not exist for these cases, and that the algebraic kernel method does not extend for them.

Theorem 4.25. *The algebraic kernel method extends for all interacting lattice walk models with small steps and group of size six if and only if* $\det M_6 = 0$*. If we let* $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$*, then we have the functional equation* $D_G(K_0\beta) = 0$*, where* K_0 *depends on the step set.*

4.2.3 Groups of Order Eight

The analysis for groups of order 8 is quite similar, so we omit most of it. Here are the results.

Model	S-polynomial
	$x_1 + \overline{x}_1 + x_1\overline{x}_2 + \overline{x}_1 x_2$
	$x_1 + \overline{x}_1 + x_1 x_2 + \overline{x}_1\overline{x}_2$

Table 4.5: Models with group of size 8.

We name the $A_1 = K_1$ and $B_1 = K_2$ iterates as before; for example, $A_2 = g_1 A_1$ and $A_3 = g_2 g_1 A_1$.

Sequence L	Alternative	Q_1^0	Q_2^0
1	2121212	$\varnothing$	1
21	121212	2	21
121	21212	12	121
2121	1212	212	121
12121	212	212	21
212121	12	12	1
1212121	2	2	$\varnothing$

Table 4.6: Reductions of group actions $g \in G$ on Q_i^0 and K_i.

The relations are the same in both models

$$A_1 = A_6\,, \quad A_2 = A_5\,, \quad A_3 = A_4\,, \quad A_7 = A_8\,, \tag{4.62}$$

$$B_1 = B_4\,, \quad B_2 = B_3\,, \quad B_4 = B_8\,, \quad B_6 = B_7\,. \tag{4.63}$$

In both cases we retrieved the same K_0.

Theorem 4.26. *The algebraic kernel method extends for all interacting lattice walk models with small steps and group of size eight. The functional equation is $D_G(K_0\beta) = 0$, where $\beta = KQ + K_{1,2}Q_{1,2}^{0,0} - \alpha$ and*

$$K_0 = A_2 A_3 B_5 B_6 = g_1(K_1) g_{21}(K_1) g_{212}(K_2) g_{1212}(K_2). \tag{4.64}$$

Chapter 5

The Obstinate Kernel Method

The next tool that we will explore is the obstinate kernel method. For its application in a two-dimensional model with interactions, see Section 4.2 of [30, p. 16]. The method is similar to the algebraic kernel method with a weighted operator, where underneath the operator notation, all we are doing is finding more valid functional equations. In both methods we apply kernel-preserving transformations on the master functional equation. However, in the obstinate kernel method we also make a substitution with one of the roots $x_k = \sigma_k(t)$ of the kernel. This is done in order to annihilate the KQ terms from the resulting equations. However, some transformations, taken together with the substitutions, do not lead to valid functional equations because some of the generating functions fail to remain power series in t after the substitution takes place. That is, one or more of the terms $g(Q_i^0)|_{x_k=\sigma_k}$ is not a power series in t for some transformation g from G and some i and k. This inspires use the following language.

Definition 5.1. *A function $f(x_1, \ldots, x_d; t)$ is admissible if it is a valid power series in the variable t, that is, there are only non-negative orders of t with nonzero coefficients. A transformation g preserves the admissibility of f with respect to a root σ_k of the kernel K if $g(f)|_{x_k=\sigma_k}$ is admissible.*

To recap, it is necessary that when we use a substitution $x_k = \sigma_k$ we only use transformations g from G that preserve the admissibility of functions gQ_i^0 for all i in the master functional equation. We will focus our work on models with groups of order four, six and eight, where we have two nontrivial generators g_1 and g_2 for the group preserving the kernel. In this case $S_i^{\pm 1} \neq 0$ for any i and choice of sign, and the roots of the kernel are

$$\sigma_k^{\pm} = \frac{1 - tS_k^0 \pm \sqrt{(1 - tS_k^0)^2 - 4t^2 S_k^1 S_k^{-1}}}{2tS_1^1}, \tag{5.1}$$

with the identities

$$\sigma_k^+ \sigma_k^- = \frac{S_k^{-1}}{S_k^1} \quad \text{and} \quad \sigma_k^+ + \sigma_k^- = \frac{1 - tS_k^0}{tS_k^1}\,. \tag{5.2}$$

If we expand the roots in series of t, we find that

$$\sigma_k^- = S_k^{-1} t + O(t^2) \quad \text{and} \quad \sigma_k^+ = \frac{1}{S_k^1 t} - \frac{S_k^0}{S_k^1} - S_k^{-1} t + O(t^2)\,, \tag{5.3}$$

where the higher order terms are the same.

Generally, only the σ_k^- substitution leads to admissible Q_i^0 terms for all i, but it also happens that σ_k^- can do this too. For an example of this, see [30, p. 11]. In this thesis we will focus on the first case and leave the σ_k^+ root analysis for future work. If we do not have enough valid equations with the σ_k^- substitution, we would have to see if the σ_k^+ substitution can help. In this thesis we will only use $\sigma = \sigma_k^-$.

It is sufficient to show that some actions of G preserve the admissibility of the functions x_1 and x_2, and thus the pair (x_1, x_2), with respect to a root, to show that the functions Q_k^0 are also admissible under those actions and substitution. That is, we only need to check when $g(x_j)|_{x_k=\sigma_k}$ is admissible some $j$and k, and for g in G. Luckily, a lot of the models with finite groups pass this test and we obtain enough valid functional equations, so that we can use the obstinate kernel method. For those cases where we find less equations, it is possible that we will be able to show that the Q_k^0 terms are power series in t via another method like in [30].

Proposition 5.2. *Consider a two-dimensional classic model with group of order four, six or eight. Models with the same group have the same admissibility properties for x_1 and x_2, and the pair (x_1, x_2). Models with group of order four all have the same admissibility properties.*

Proof. All we need to do is compute the actions on the functions and check if substituting σ_k^- leads to admissible ones. This is tedious to do by hand but straight-forward to do on a computer. The results are summarized in tables in the following sections. However, we can perform the analysis for groups of order four without difficulty.

The transformations 1 and g_2 leave the function x_1 unchanged, but the transformations g_1 and $g_1 g_2$ change it to $g_1(x_1) = \overline{x}_1 S_1^{-1}/S_1^1$. Note that $\overline{\sigma}_1 = \sigma_1^+ S_1^1/S_1^{-1}$, so $g_1(x_1)|_{x_1=\sigma_1} = \sigma_1^+$ and this function is not admissible. Therefore x_1's admissibility with respect to σ_1 is preserved only by the transformations $\{1, g_2\}$.

Since models with groups of order four need to have one symmetry across an axis, without loss of generality, we can assume the symmetry is across the x_1 axis. That is, $g_2(x_2) = \overline{x}_2$, and the

$x_1 = \sigma_1$ substitution trivially leads to an admissible function. Therefore, all of G preserves the admissibility of x_2 with respect to σ_1 and finally, only 1 and g_2 preserve the admissibility for both x_1 and x_2 with respect to σ_1. The analysis is analogous for the σ_2 substitution. □

5.1 Initial Computations

This classification shows that we can treat all of the models with group of order four with the same method. The situation is somewhat more complicated when the order of the group is six or eight, as we shall see in the following sections. Suppose that we have a set of valid equations for a substitution $x_k = \sigma_k$. They are a linear system in the unknown functions $Q^{0,0}_{1,2}$, Q^0_1, Q^0_2, and their orbits under the set of valid group actions. We hope for a high level of redundancy in these equations, because it will allow us to eliminate some of the unknowns by taking a weighted sum over the valid equations. That is, we have a weighted orbital summation operator paired with the substitution $x_k = \sigma_k$. It turns out that it is enough to have $|G|/2$ transformations preserving the admissibility for both x_1 and x_2.

5.1.1 Groups of Order Four

We start with the admissibility table for groups of order four.

Root	$x_1(t)$	$x_2(t)$	$x_1(t)$ and $x_2(t)$
$\sigma_1(t)$	1, g_2	All	1, g_2
$\sigma_2(t)$	All	1, g_1	1, g_1

Table 5.1: Admissibility-preserving transformations with respect to roots of the kernel with group of order four.

The σ substitution lead to essentially the same analysis, so we will only work with the σ_1 substitution, together with actions 1 and g_2. Let F^* denote $g_2(F)$. The functional equations corresponding the the group actions are

$$KQ - K_1Q^0_1 - K_2Q^0_2 + K_{12}Q^{0,0}_{1,2} = \alpha\,, \tag{5.4}$$

$$KQ^* - K^*_1Q^{0,*}_1 - K^*_2Q^0_2 - K^*_{12}Q^{0,0}_{1,2} = \alpha\,. \tag{5.5}$$

Once we make the substitution, the function Q^0_2 will be difficult to expand, so we would like to eliminate it first. We multiply the equations by K^*_2 and K_2, respectively, and subtract the second

one from the first one. The resulting equation is

$$K_2^* KQ - K_2 KQ^* - (K_2^* K_1 Q_1^0 - K_2 K_1^* Q_1^{0,*}) + (K_2^* K_{12} - K_2 K_{12}^*) Q_{1,2}^{0,0} - \alpha(K_2^* - K_2) = 0\,, \tag{5.6}$$

where we can set $x_1 = \sigma_1$ to eliminate the K term. The substitution will be implicit in the notation from now on. We have

$$(K_2^* K_1 Q_1^0 - K_2 K_1^* Q_1^{0,*}) - (K_2^* K_{12} - K_2 K_{12}^*) Q_{1,2}^{0,0} + \alpha(K_2^* - K_2) = 0. \tag{5.7}$$

We can recast this elimination via the weighted orbital operator

$$D_2 = g_2(K_2) \cdot 1 - K_2 g_2 = (1 - g_2) K_2^*\,. \tag{5.8}$$

If we apply it on the master equation, we obtain

$$\begin{aligned} D_2(KQ - K_1 Q_1^0 - K_2 Q_2^0 + K_{12} Q_{1,2}^{0,0}) = 0\,, \\ D_2(KQ) - D_2(K_1 Q_1^0) + D_2(K_{12} Q_{1,2}^{0,0}) = 0\,, \end{aligned} \tag{5.9}$$

and once we make the σ_1 substitution, we find that

$$D_2(K_1 Q_1^0) - D_2(K_{12}) Q_{1,2}^{0,0} + D(\alpha) = 0 \tag{5.10}$$

and this is the same as Equation 5.6. The following notation will be useful in the next few sections.

Definition 5.3. *Let L be a concatenated list, $\Delta_L = 1 - g_L$ and let X be some function. The operator $D_L[X]$ is defined by its action on functions F as*

$$D_L[X](F) = \Delta_L(XF) = (1 - g_L)(XF) = XF - g_L(XF)\,. \tag{5.11}$$

We will drop the X from the notation if it is clear from context. In particular, $D_L[1] = \Delta_L$.

For example, $D_2 = D_2[K_2^*] = D_2[g_2(K_2)]$.

Example 5.4. If we use $g_L(X)$ instead of X in the definition above, we obtain the Ore operator

$$D_L[g_L(X)] = g_L(X) - X g_L\,. \tag{5.12}$$

Now, we can expand some of the terms in the functional equation. First,

$$\begin{aligned} D_2(\alpha) &= \alpha(K_2^* - K_2) \\ &= \alpha((c_2 - tx_2\overline{\mu}_2 S_2^{-1}) - (c_2 - t\overline{x}_2 S_2^{-1})) \\ &= \alpha t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))S_2^{-1}(\sigma_1)\,. \end{aligned} \tag{5.13}$$

Then,

$$\begin{aligned} D_2(K_{12}) &= K_2^* K_{12} - K_2 K_{12}^* \\ &= c_1 c_2 t(\overline{x}_2 - x_2\overline{\mu}_2)S_2^{-1} - c_2 t\overline{\sigma}_1(\overline{x}_2 - x_2\overline{\mu}_2)S_{1,2}^{-1,-1} \\ &= c_2 t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))(c_1 S_2^{-1}(\sigma_1) - \overline{\sigma}_1 S_{1,2}^{-1,-1})\,. \end{aligned} \tag{5.14}$$

Proposition 5.5. *For models with group of order four,*

$$D_2(K_1 Q_1^0) - D_2(K_{12})Q_{1,2}^{0,0} + D(\alpha) = 0\,, \tag{5.15}$$

where

$$D_2(\alpha) = \alpha t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))S_2^{-1}(\sigma_1)\,, \tag{5.16}$$

$$D_2(K_{12}) = c_2 t(\overline{x}_2 - x_2\overline{\mu}_2(\sigma_1))(c_1 S_2^{-1}(\sigma_1) - \overline{\sigma}_1 S_{1,2}^{-1,-1})\,. \tag{5.17}$$

If the model is highly-symmetric, we see that $g_2 S_1^{-1} = S_1^{-1}$ and therefore, $g_2 K_1 = K_1$, and we may bring that factor outside of the operator, so $D_2(K_1^* F) = K_1 D_2(F)$ for functions F. It turns out that this happens for both substitutions if and only if the model is highly-symmetric, meaning that the analysis will be easier only in those cases. This is easily verified using a computer since there are not that many models with group of order four. We will investigate the highly-symmetric models in more detail in Section 5.2.

Proposition 5.6. *Consider a classic model with a group of order four. If $i \neq j$, then $D_i[g_i K_j] = K_j \Delta_i$ in the σ_j substitutions if and only if the model has a symmetry across the x_i axis. Thus, we find that the K functions factor out of the operators in both substitutions if and only if the model is highly-symmetric.*

5.1.2 Groups of Order Six

We start with the admissibility tables for the two groups of order six.

Root	x_1	x_2	(x_1, x_2)
σ_1	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_2, g_{21}$	$1, g_2, g_{21}$
σ_2	$1, g_1, g_2, g_{12}$	$1, g_1, g_{12}, g_{121}$	$1, g_1, g_{12}$

Table 5.2: Admissibility-preserving transformations with respect to roots of the kernel with group of order six for the models $S11$, $S12$, $S13$.

Root	x_1	x_2	(x_1, x_2)
σ_1	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_{12}, g_{121}$	$1, g_{121}$
σ_2	$1, g_2, g_{21}, g_{121}$	$1, g_1, g_{12}, g_{121}$	$1, g_{121}$

Table 5.3: Admissibility-preserving transformations with respect to roots of the kernel with group of order six for the models $S21$, $S22$, $S33$.

We focus on the first three models because they are amenable to the straight-forward method. The latter three would need more work, since there are too few valid actions.

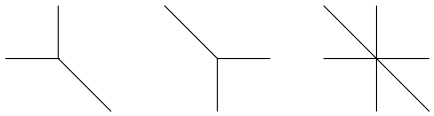


Figure 5.1: Steps for models $S11$, $S12$, $S13$.

For consistency, we will work with σ_1 where the actions are 1, g_2, g_{21}. From the previous chapter, we know these actions reduce on the function Q_i^0.

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21

Table 5.4: Reductions of group actions $g \in G$ on Q_1^0 and Q_2^0.

As in the previous section, we wish to eliminate $X_2 = g_2 Q_1^0$ and $Y = Q_2^0$ because they would

have unwieldy arguments once the substitution is made. Let

$$\beta = K_{12}Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1)\,, \tag{5.18}$$

so that we have

$$AX + BY = \beta\,, \tag{5.19}$$

$$A_2 X_2 + B_2 Y = \beta_2\,, \tag{5.20}$$

$$A_{21} X + B_{21} Y_{21} = \beta_{21}\,. \tag{5.21}$$

We want a linear combination of β, β_2 and β_{21} that eliminates X_2 and Y. In this case, we may write

$$\begin{aligned} a\beta - b\beta_2 + c\beta_{21} &= aAX + (aB - bB_2)Y + (cA_{21} - bA_2)X_2 + cB_{21}Y_{21} &(5.22)\\ &= aAX + cB_{21}Y_{21}\,, &(5.23)\end{aligned}$$

and for the elimination to work, we need $aB = bB_2$ and $cA_{21} = bA_2$. One solution for a, b and c without denominators is **[fix: Check above]**

$$a = A_{21}B_2\,, \quad b = A_{21}B\,, \quad c = A_2 B\,. \tag{5.24}$$

The elimination procedure yields

$$\begin{aligned} AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 &= A_{21}B_2\beta - A_{21}B\beta_2 + A_2B\beta_{21}\\ &= (A_{21}B_2 - A_{21}B + A_2B)\beta \end{aligned} \tag{5.25}$$

and we can recast the whole process equation into operator notation.

Proposition 5.7. *Let*

$$D(F) = A_{21}B_2F - A_{21}Bg_2(F) + A_2Bg_{21}(F)\,. \tag{5.26}$$

Then

$$D(AX + BY) = AA_{21}B_2Q_1^0 + A_2BB_{21}g_{21}Q_2^0 = \beta D(1)\,. \tag{5.27}$$

For each case, we can compute $D(1)$, but since the expressions are somewhat complicated and contain a square root, we will not list them here. Terms $AA_{21}B_2$ and A_2BB_{21} contain roots too.

5.1.3 Groups of Order Eight

We start with the admissibility tables for the two groups of order eight.

Root	x_1	x_2	(x_1, x_2)
σ_1	1, g_2, g_{21}, g_{212}	All	1, g_2, g_{21}, g_{212}
σ_2	1, g_1, g_2, g_{12}, g_{121}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_1, g_{12}, g_{121}

Table 5.5: Admissibility-preserving transformations with respect to roots of the kernel with group of order eight for model $S1$.

Root	x_1	x_2	(x_1, x_2)
σ_1	1, g_2, g_{21}, g_{212}	All	1, g_2, g_{21}, g_{212}
σ_2	1, g_2, g_{21}, g_{121}, g_{212}, g_{2121}	1, g_1, g_{12}, g_{121}	1, g_{121}

Table 5.6: Admissibility-preserving transformations with respect to roots of the kernel with group of order eight for model $S2$.

As in the previous section, we will work with the model with all the necessary actions. In this case, the actions are 1, g_2, g_{21}, g_{212}, and the step polynomial is

$$S = \overline{x}_1 + x_1 + \overline{x}_1 x_2 + x_1 \overline{x}_2 \,. \tag{5.28}$$

Note that the model here is equivalent to the one seen in the model in [30], because they have the same step set and master functional equation, so the analysis will be the same. Using our notation and operators, we will be able to redo some of the authors' derivations in a cleaner way. First, we have the table of reductions for group action in this model.

Sequence L	Q_1^0	Q_2^0
$\varnothing$	$\varnothing$	$\varnothing$
2	2	$\varnothing$
21	2	21
212	212	21

Table 5.7: Reductions of group actions that preserve admissibility with respect to σ_1.

Similarly to previous sections, let $\beta = K_{12} Q_{1,2}^{0,0} - \alpha = \alpha(b_1 b_2 Q_{1,2}^{0,0} - 1)$ and $X = Q_1^0$ and

$Y = Q_2^0$. Note that β is left unchanged by group actions. We have the four equations

$$AX + BY = \beta\,, \tag{5.29}$$

$$A_2X_2 + B_2Y = \beta\,, \tag{5.30}$$

$$A_{21}X_2 + B_{21}Y_{21} = \beta\,, \tag{5.31}$$

$$A_{212}X_{212} + B_{212}Y_{21} = \beta\,, \tag{5.32}$$

where we wish to eliminate the functions X_2, Y and Y_{21} because they contain σ_1 as part of their arguments. To do this, we first use the first two and latter two equations to eliminate Y and Y_{21}, respectively. That is, we compute B_2 times the first equation minus B times the second equation, and B_{212} times the third equation minus B_{21} from the last equation. Then we use the resulting two equations to eliminate X_2. The first step of the procedure is the same as applying the following two operators on the master functional equation and substituting $x_1 = \sigma_1$. They are

$$D_2[B_2] = B_2 - Bg_2\,, \tag{5.33}$$

$$-g_{212}D_2[B_2] = g_{212}\,Bg_2 - g_{212}\,B_2 = B_{212}\,g_{21} - B_{21}\,g_{212}\,, \tag{5.34}$$

and we obtain the equations

$$AB_2X - A_2BX_2 = \beta(B_2 - B)\,, \tag{5.35}$$

$$A_{21}B_{212}X_2 - A_{212}B_{21}X_{212} = \beta(B_{212} - B_{21})\,. \tag{5.36}$$

To eliminate X_2, we multiply the equations by $A_{21}B_{212}$ and A_2B, add them together and obtain

$$A_{21}B_{212}AB_2X - A_2BA_{212}B_{21}X_{212} = \beta(A_{21}B_{212}(B - B_2) + A_2B(B_{212} - B_{21}))\,. \tag{5.37}$$

This final equation is equivalent to applying the following operator on the master functional equation:

$$\begin{aligned} D &= A_{21}B_{212}D_2[B_2] - A_2Bg_{212}D_2[B_2] \\ &= (A_{21}B_{212} - A_2Bg_{212})D_2[B_2] \\ &= (A_{21}B_{212} - g_{212}A_{21}B_{212})D_2[B_2] \\ &= D_{212}[A_{21}B_{212}]D_2[B_2]\,. \end{aligned} \tag{5.38}$$

From the master functional equation, we have

$$D(\beta) = \beta D_{212}[A_{21}B_{212}](B_2 - B)\,, \tag{5.39}$$

$$\begin{aligned} D(AX + BY) &= D(AX) \\ &= D_{212}[A_{21}B_{212}]D_2[B_2](AX) \\ &= D_{212}[A_{21}B_{212}](AB_2X - A_2BX_2) \\ &= D_{212}[A_{21}B_{212}](AB_2X) \\ &= D_{212}[AA_{21}B_2B_{212}](X)\,. \end{aligned} \tag{5.40}$$

Note that $A_{212} = A$, so we can factor out A from the operator. We have the following result.

Proposition 5.8. *For the first model with group of order eight, we have*

$$AD_{212}[A_{21}B_2B_{212}]\left(Q_1^0\right) = \beta D_{212}[A_{21}B_{212}](B_2 - B)\,, \tag{5.41}$$

where $A_L = g_L(K_1)$, $B_L = g_L(K_2)$ and $\beta = \alpha(b_1b_2Q_{1,2}^{0,0} - 1)$, and the equality is valid only once we set $x_1 = \sigma_1$ in this expression.

For the analysis in the σ_2 substitution, we switch the A's and the B's, and the 1's and the 2's and obtain an analogous statement.

It was found in [30] that the coefficients in front of the X and X_{212} terms become Laurent polynomials once we do the σ_k substitutions. The coefficient in front of β can be reduced to a rational function plus a rational function times a square root after properly conjugating. The calculations are exceedingly complicated and would be difficult to do by hand. The square root part does not depend on a_2 in the first case, and a_1 in the second case. This observation allows the authors to find a relationship between the interacting and non-interacting versions of $Q_{1,2}^{0,0}$.

5.2 Highly-symmetric Two-dimensional Models

For highly-symmetric models, the first equation in Proposition 5.5 has the simplified form

$$x_1 = \sigma_1 : \quad \Delta_2(K2^*Q_1^0) = \frac{1}{K_1}\left(\Delta_2(K_2^*K_{12})Q_{1,2}^{0,0} - \alpha(K_2^* - K_2)\right)\,, \tag{5.42}$$

$$x_2 = \sigma_2 : \quad \Delta_1(K1^\bullet Q_2^0) = \frac{1}{K_2}\left(\Delta_1(K_1^\bullet K_{12})Q_{1,2}^{0,0} - \alpha(K_1^\bullet - K_1)\right)\,. \tag{5.43}$$

It is enough to perform the analysis for one of the equations, since only the model with two horizontal or vertical steps removed requires working with both equations. In highly-symmetric models,

$\mu_2 = S_2^{-1}/S_2^1 = 1$ and

$$\alpha(K_2^* - K_2) = \alpha t(\overline{x}_2 - x_2) S_2^{-1}(\sigma_1)\,, \tag{5.44}$$

$$\Delta_2(K_2^* K_{12}) = c_2 t(\overline{x}_2 - x_2)(c_1 S_2^{-1}(\sigma_1) - \overline{\sigma}_1 S_{1,2}^{-1,-1})\,. \tag{5.45}$$

To compute $\Delta_2(K2^*Q_1^0)$, we need to compute $K_2(\sigma_1, x_2)$ and $K_2^*(\sigma_1, x_2)$ given by

$$K_2 = c_2 - t\overline{x}_2 S_2^{-1}(\sigma_1) \quad \text{and} \quad K_2^* = c_2 - t x_2 S_2^{-1}(\sigma_1)\,, \tag{5.46}$$

where we can find that $S_2^{-1}(\sigma)$ is a rational function given by

$$S_2^{-1}(\sigma_1) = \frac{S_{1,2}^{1,1} + t S_{1,2}^{0,1} S_{1,2}^{1,0}}{t\left(S_{1,2}^{1,0} + S_{1,2}^{1,1}(\overline{x}_2 + x_2)\right)}\,. \tag{5.47}$$

Here is a table summarizing the values of this function for the different models.

Model	$S_{1,2}^{0,1}$	$S_{1,2}^{1,0}$	$S_{1,2}^{1,1}$	$S_2^{-1}(\sigma_1)$
	1	1	0	1
	0	0	1	$\overline{t}/(\overline{x}_2 + x_2)$
	1	0	1	$\overline{t}/(\overline{x}_2 + x_2)$
	0	1	1	$\overline{t}/(\overline{x}_2 + 1 + x_2)$
	1	1	1	$(1 + \overline{t})/(\overline{x}_2 + 1 + x_2)$

Table 5.8: Function $S_2^{-1}(\sigma_1)$ for highly-symmetric models.

Lemma 5.9. *For highly-symmetric models with $e = S_{1,2}^{1,1}$, $f = S_{1,2}^{1,0}$ and $g = S_{1,2}^{0,1}$, we have*

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(e + tfg)}{S_1^1 K_1}\,, \tag{5.48}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(e + tfg) - et\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)}\,. \tag{5.49}$$

5.2.1 Cartesian Model – Steps N, S, E, W

This is the simplest case to solve since $S_2^{-1}(\sigma_1) = 1$. We will essentially do the same calculations as we did after Proposition 2.6 in Chapter 2 but for two dimensions. This time we will not obtain a linear relationship between the quantities. Afterwards, we will solve for Q in terms of various version of itself with partial interactions, and $Q_{1,2}^{0,0}$.

Here $K_{12} = c_1 c_2$, so it behaves like α and we have

$$\Delta_2(K2^* Q_1^0) + \frac{\alpha\beta(K_2^* - K_2)}{K_1} = 0\,, \tag{5.50}$$

where $\beta = 1 - b_1 b_2 Q_{1,2}^{0,0}$ and

$$\frac{\alpha\beta(K_2^* - K_2)}{K_1} = \frac{\alpha\beta t(\overline{x}_2 - x_2)}{K_1} = \alpha\beta f(c_1, x_2)\,, \tag{5.51}$$

since K_1 does not depend on c_2 or x_1. The functional equation becomes

$$(c_2 - t x_2) Q_1^0(x_2) - (c_2 - t\overline{x}_2) Q_1^0(\overline{x}_2) + \alpha\beta f = 0\,. \tag{5.52}$$

We extract the x_2 and $x_2 b_2^0$ terms to find the equations

$$c_2 Q_{1,2}^{0,1} - t Q_{1,2}^{0,0} + \alpha\beta f_2^1 = 0\,, \tag{5.53}$$

$$-t\check{Q}_{1,2}^{0,0} + \overline{a}_1 f_2^1 = 0\,. \tag{5.54}$$

We can substitute the latter equation in the first one to obtain the identity

$$b_2 Q_{1,2}^{0,1} = t(a_2 Q_{1,2}^{0,0} - \beta\check{Q}_{1,2}^{0,0})\,. \tag{5.55}$$

In general, we have the following result. The analysis is similar to the one we will see in Corollary 5.13, so we omit it.

Lemma 5.10. *For non-negative integers m and n,*

$$b_2 Q_{1,2}^{m,n+1} = t(a_2 Q_{1,2}^{m,n} - \beta\check{Q}_{1,2}^{m,n})\,, \tag{5.56}$$

$$b_1 Q_{1,2}^{m+1,n} = t(a_1 Q_{1,2}^{m,n} - \beta\widehat{Q}_{1,2}^{m,n})\,. \tag{5.57}$$

From the first equation we have

$$b_2 Q^{1,1}_{1,2} = t(a_2 Q^{1,0}_{1,2} - \beta \breve{Q}^{1,0}_{1,2})\,. \tag{5.58}$$

If we multiply it by b_1 and use the relations in the lemma, we find that

$$\begin{aligned} b_1 b_2 Q^{1,1}_{1,2} &= t a_2 t(a_1 Q^{0,0}_{1,2} - \beta \widehat{Q}^{0,0}_{1,2}) - t\beta t(a_1 \breve{Q}^{0,0}_{1,2} - \beta \breve{\widehat{Q}}^{0,0}_{1,2}) \\ &= t^2(a_1 a_2 Q^{0,0}_{1,2} - a_2 \beta \widehat{Q}^{0,0}_{1,2} - a_1 \beta \breve{Q}^{0,0}_{1,2} + \beta^2 \breve{\widehat{Q}}^{0,0}_{1,2})\,. \end{aligned} \tag{5.59}$$

Let $G = Q^{0,0}_{1,2}$, $\widehat{G} = G_1$ and $\breve{G} = G_2$. From the $x_1^0 x_2^0$ term of the master equation, we have

$$Q^{0,0}_{1,2} = 1 + t a_1 a_2 (Q^{1,0}_{1,2} + Q^{0,1}_{1,2})\,. \tag{5.60}$$

Multiplying it by $b_1 b_2$ and using the relations above, leads to

$$\begin{aligned} b_1 b_2 G &= b_1 b_2 + t a_1 a_2 (b_2 t(a_1 G - \beta G_1) + b_1 t(a_2 G - \beta G_2) \\ &= b_1 b_2 + t^2 a_1 a_2 (b_2 a_1 G + b_1 a_2 G - b_2 \beta G_1 - b_1 \beta G_2)\,. \end{aligned} \tag{5.61}$$

Using the relation for β and simplifying the expression leads to

$$G = \frac{b_1 b - 2 - t^2 a_1 a_2 (b_2 G_1 + b_1 G_2)}{b_1 b_2 - t^2 a_1 a_2 (a_2 b_1 + a_1 b_2 + b_1 b_2 (b_2 G_1 + b_1 G_2))}\,. \tag{5.62}$$

If we let $F = (b_2 G_1 + b_1 G_2)$ and perform the long division with respect to F, we find that

$$Q^{0,0}_{1,2} = \frac{1}{b_1 b_2} \left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)} \right)\,. \tag{5.63}$$

To find an equation for Q we need to first take the second set of equations in Lemma 5.10, multiply them by x_2^n and sum them up. The resulting equations are

$$b_1 Q^{m+1}_1 = t(a_1 Q^m_1 - \beta \widehat{Q}^m_1) \tag{5.64}$$

and we can sum them with respect to x_1 to find

$$b_1 (Q - Q^0_1) = x_1 t(a_1 Q - \beta \widehat{Q})\,. \tag{5.65}$$

We can solve this it for Q_1^0 as obtain

$$b_1 Q_1^0 = (b_1 - a_1 x_1 t)Q + x_1 t \beta \widehat{Q}\,. \tag{5.66}$$

We substitute this expression, along with an analogous one for Q_2^0, into the original functional equation, and after some algebra, we find that

$$Q = \frac{(1 - b_1 b_2 G)(\alpha + t x_1 \bar{b}_1 K_1 \widehat{Q} + t x_2 \bar{b}_2 K_2 \check{Q})}{K - (1 - \bar{c}_1 x_1 t)K_1 - (1 - \bar{c}_2 x_2 t)K_2} \tag{5.67}$$

We recognize some of the quantities as modified K_i terms.

Theorem 5.11. *The Cartesian model in two dimensions has generating functions satisfying the equations*

$$Q_{1,2}^{0,0} = G = \frac{1}{b_1 b_2}\left(1 - \frac{b_1 b_2 (1 - b_1 b_2) - t^2 a_1 a_2 (a_1 b_2 + a_2 b_1)}{b_1 b_2 + t^2 a_1 a_2 (a_1 b_2 + a_2 b_1 + b_1 b_2 F)}\right)\,, \tag{5.68}$$

where $F = (b_2 \widehat{G} + b_1 \check{G})$*, and*

$$Q = \frac{(1 - b_1 b_2 Q_{1,2}^{0,0})(\alpha - \bar{b}_1 K_1 \widehat{K}_1^{\bullet} \widehat{Q} - \bar{b}_2 K_2 \check{K}_2^{*} \check{Q})}{K - \bar{c}_1 K_1 K_1^{\bullet} - \bar{c}_2 K_2 K_2^{*}}\,. \tag{5.69}$$

This shows that Q is built from versions of itself, each with a single interaction, and $Q_{1,2}^{0,0}$.

5.2.2 Diagonal Cartesian Model – Steps NE, NW, SE, SW

This model is similar to the one covered in [21], where we can find a model of two interacting walks confined in a semi-infinite strip. The models are different because there is an extra term in their functional equation. We will go through calculations similar to the ones in the previous subsection and we will find that $Q_{1,2}^{0,0}$ has a linear relationship with its partially-marked versions. However we will not find an identity for Q.

In this case,

$$\frac{\alpha(K_2^{*} - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)}{K_1(\overline{x}_2 + x_2)}\,, \tag{5.70}$$

$$\frac{\Delta_2(K_2^{*} K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1 - t\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)} = \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2}\,, \tag{5.71}$$

and the functional equation becomes

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x}_2 - x_2)}{\overline{x}_2 + x_2} Q_{1,2} + \alpha \frac{\overline{x}_2 - x_2}{(\overline{x}_2 + x_2)K_1} = 0\,. \tag{5.72}$$

Surprisingly enough, the only remaining term depending on σ_1 is K_1, so it will be of no consequence to the rest of the analysis. We just need to take care of the remaining terms and we do this by multiplying the equation by $\overline{x}_2 + x_2$. We have

$$R^* = (\overline{x}_2 + x_2)K_2^* = c_2(\overline{x}_2 + x_2) - x_2 = c_2\overline{x}_2 - \overline{a}_2 x_2\,, \tag{5.73}$$

$$R = (\overline{x}_2 + x_2)K_2 = c_2(\overline{x}_2 + x_2) - \overline{x}_2 = c_2 x_2 - \overline{a}_2\overline{x}_2\,. \tag{5.74}$$

The R functions satisfy the identities

$$R^* - R = v(K_2^* - K_2) = \overline{x}_2 - x_2\,, \tag{5.75}$$

$$\overline{x}_2 R^* - x_2 R = \overline{x}_2 c_2 v - 1 - x_2 c_2 v + 1 = c_2 v(\overline{x}_2 - x_2)\,. \tag{5.76}$$

Finally, the functional equation becomes

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{0,0} + \overline{a}_2\delta_0 = 0\,, \tag{5.77}$$

where $\delta_0 = \overline{a}_1(\overline{x}_2 - x_2)\overline{K}_1$.

Next, we will work towards finding equations like those in Lemma 5.10 but for diagonal steps. To do that, we need to find higher order equations like 5.77 above. We begin our analysis with the master functional equation for diagonal steps:

$$(1 - tS)Q - (c_1 - tv\overline{x}_1)Q_1^0 - (c_2 - t\overline{x}_2 S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x}_1\overline{x}_2)Q_{1,2}^{0,0} = \overline{a}_1\overline{a}_2\,. \tag{5.78}$$

Upon extracting the x_1^0 coefficient, we find that

$$Q_1^0 = \overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1\,. \tag{5.79}$$

We can extract higher order coefficients too. Let $k \geqslant 1$. Then the x_2^k coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0 Q_1^k - tS_1^1 Q_1^{k-1} - c_2 Q_{1,2}^{k,0} \\ &\quad + t\overline{x}_2(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2 Q_{1,2}^{k,0} + t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \end{aligned} \tag{5.80}$$

We can rewrite the last one in a form more convenient for the next few steps:

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2 Q_{1,2}^{k,0} - t\overline{x}_2(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\,. \tag{5.81}$$

We have

$$Q_1^0 = \overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1\,, \tag{5.82}$$

$$Q_1^{0,*} = \overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}\,. \tag{5.83}$$

Combining them with Equation 5.77 gives

$$\begin{aligned}
0 &= R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x}_2 - x_2) Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\
&= R^*(\overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a}_2 + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) \\
&\quad - c_2(\overline{x}_2 - x_2) Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a}_2 + c_2 Q_{1,2}^{0,0}) \\
&\quad - c_2(\overline{x}_2 - x_2) Q_{1,2}^{0,0} + \overline{a}_2\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a}_2((\overline{x}_2 - x_2) + \delta_0)\,,
\end{aligned} \tag{5.84}$$

and we have the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a}_2\delta_1 = 0\,, \tag{5.85}$$

where

$$\delta_1 = \frac{(\overline{x}_2 - x_2) + \delta_0}{a_1 tv}\,. \tag{5.86}$$

This equation together with Equation 5.77 form the base cases for the following result.

Lemma 5.12. *Let m be an integer greater than zero. Then*

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x}_2 - x_2) Q_{1,2}^{m,0} + \overline{a}_2\delta_m = 0\,, \tag{5.87}$$

where δ_m is not a function of a_2.

Proof. We prove this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$, since we cannot have walks ending on $(1, 0)$. Assume that the

formula is true for all cases up to and including m. Then we use Equation 5.81 to calculate

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*} \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*}) \\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x}_2R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x}_2 - x_2)Q_{1,2}^{m-1,0} - \overline{a}_2\delta_{m-1}) \\
&+ c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x}_2 - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} \\
&- c_2tv(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0}) - \overline{a}_2tv\delta_{m-1}) \,.
\end{aligned} \tag{5.88}$$

Combining this expression with the equation from the lemma leads to

$$0 = R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x}_2 - x_2)Q_{1,2}^{m+1,0} + \overline{a}_2\delta_{m+1} \,, \tag{5.89}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1} \tag{5.90}$$

is independent from a_2. □

Of course, there is an analogous statement for Q_2^n if we follow the same analysis with the variable names switched. We can expand R^* and R to a form we can use:

$$\begin{aligned}
0 = &(c_2\overline{x}_2 - \overline{a}_2x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a}_2\overline{x}_2)Q_1^m(\overline{x}_2) \\
&- c_2(\overline{x}_2 - x_2)Q_{1,2}^{m,0} + \overline{a}_2\delta_m \,.
\end{aligned} \tag{5.91}$$

We extract the x_2 coefficient

$$\begin{aligned}
c_2Q_{1,2}^{m,2} - \overline{a}_2Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2\delta_m = 0 \\
c_2Q_{1,2}^{m,2} - \overline{a}_2Q_{1,2}^{m,0} + [x_2^1]\overline{a}_2\delta_m = 0 \,.
\end{aligned} \tag{5.92}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 1$.

$$c_2Q_{1,2}^{m,n+2} - \overline{a}_2Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a}_2\delta_m = 0 \,. \tag{5.93}$$

This formula also contains the $n = 0$ case, so we can group them together. Set $a_2 = 1$ to find

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0 \,, \tag{5.94}$$

and substitute the resulting expression into the equation above to obtain

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n} \,. \tag{5.95}$$

We finally have the following result.

Corollary 5.13. *For integers m and n greater than zero, we have the following relations*

$$b_1 Q_{1,2}^{m+2,n} = Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n} \,, \tag{5.96}$$

$$b_2 Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n} \,. \tag{5.97}$$

We can sum up those equations to retrieve the following functional equations.

Corollary 5.14. *For integers m greater than zero, we have the following identities*

$$Q_1^m - \widehat{Q}_1^m = b_1 Q_1^{m+2} \,, \tag{5.98}$$

$$x_1^2 (Q_2^n - \widehat{Q}_2^n) = b_1 (Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n}) \,, \tag{5.99}$$

$$x_1^2 (Q - \widehat{Q}) = b_1 (Q - x_1 Q_1^1 - Q_1^0) \,, \tag{5.100}$$

where we may also set $a_2 = 1$ to obtain further identities.

[Bonus: Check to see if we can find an equation for Q here.] Now we can move onto the main result about the diagonal walks — the relation between $Q_{1,2}^{0,0}$ and its non-interacting versions. We use the master equation to extract the $x_1^0 x_2^0$ and $x_1^1 x_2^1$ coefficients and obtain the following identities:

$$Q_{1,2}^{0,0} = 1 + t a_1 a_2 Q_{1,2}^{1,1} \tag{5.101}$$

$$Q_{1,2}^{1,1} = t (Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2}) \,. \tag{5.102}$$

We also have

$$b_1 b_2 Q_{1,2}^{2,2} = b_1 (Q_{1,2}^{2,0} - \breve{Q}_{1,2}^{2,0}) = Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0} \,, \tag{5.103}$$

together with

$$b_1 b_2 Q_{1,2}^{2,0} = b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \,, \tag{5.104}$$

$$b_1 b_2 Q_{1,2}^{0,2} = b_1 Q_{1,2}^{0,0} - b_1 \breve{Q}_{1,2}^{0,0} \,. \tag{5.105}$$

Furthermore,

$$\begin{aligned} b_1b_2Q_{1,2}^{1,1} &= t(b_1b_2Q_{1,2}^{0,0} + b_1b_2Q_{1,2}^{2,0} + b_1b_2Q_{1,2}^{0,2} + b_1b_2Q_{1,2}^{2,2}) \\ &= t((b_1b_2 + b_1 + b_2 + 1)Q_{1,2}^{0,0} - (b_2+1)\widehat{Q}_{1,2}^{0,0} - (b_1+1)\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}) \\ &= t(a_1a_2Q_{1,2}^{0,0} - a_2\widehat{Q}_{1,2}^{0,0} - a_1\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.106}$$

Our final identity is

$$\begin{aligned} b_1b_2Q_{1,2}^{0,0} &= b_1b_2 + ta_1a_2b_1b_2Q_{1,2}^{1,1} \\ &= b_1b_2 + t^2a_1a_2(a_1a_2Q_{1,2}^{0,0} - a_2\widehat{Q}_{1,2}^{0,0} - a_1\breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0})\,. \end{aligned} \tag{5.107}$$

As before, let G stand for $Q_{1,2}^{0,0}$ and G_1, G_2 and $G_{1,2}$ for the various extractions of b_1^0 and b_2^0 coefficients. The relation becomes

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})\,, \tag{5.108}$$

or written as a linear combination,

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2\,. \tag{5.109}$$

We can divide by a_1a_2 and obtain the equation

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2\,, \tag{5.110}$$

Theorem 5.15. *$B(fQ_{1,2}^{0,0}) = c_1c_2$, where $f(a_1,a_2,t) = c_1c_2 - a_1a_2t^2$ and $Q_{1,2}^{0,0}$ is the generating function for walks ending on the origin in the Diagonal Cartesian model.*

It may be possible to find a similar combination for the function Q but we will not do this here.

5.2.3 The Full Model – All Steps

We will finally take start the analysis for the model featuring all the steps. The situation is more difficult because we will no longer have a rational function in front of the $Q_{1,2}^{0,0}$ term and we are not

sure how to proceed from there. We start with the following computation:

$$\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{S_1^1 K_1}\,, \tag{5.111}$$

$$\frac{\Delta_2(K_2^* K_{12})}{K_1} = \frac{c_2(\overline{x}_2 - x_2)(c_1(1+t) - t\overline{\sigma}_1 S_1^1)}{S_1^1(c_1 - t\overline{\sigma}_1 S_1^1)} \tag{5.112}$$

$$= \frac{c_2(\overline{x}_2 - x_2))}{S_1^1}\left(1 - \frac{t}{c_1 - t\overline{\sigma}_1 S_1^1}\right)\,. \tag{5.113}$$

We will need to do some more work to find a representation for that last fraction. For that, we need the σ_1 roots given by

$$\sigma_1 = \frac{u - \sqrt{u^2 - 4w^2}}{2w} \quad \text{and} \quad \overline{\sigma}_1 = \frac{u + \sqrt{u^2 - 4w^2}}{2w} \tag{5.114}$$

for $u = 1 - tv$ and $w = t(1+v)$. Then we have

$$\begin{aligned} \frac{t}{c_1 - \overline{\sigma}_1 w} &= \frac{2t}{2c_1 - u - \sqrt{u^2 - 4w^2)}} \\ &= \frac{t(2c_1 - u + \sqrt{u^2 - 4w^2})}{2(c_1^2 - c_1 u + w^2)}\,. \end{aligned} \tag{5.115}$$

Let $v = \overline{x}_2 + x_2$. Then the denominator has the term

$$c_1^2 - c_1 u + w^2 = c_1^2 - c_1(1 - tS_1^0) + t^2(S_1^1)^2 \tag{5.116}$$

$$= \overline{a}_1^2 b_1 - \overline{a}_1 b_1 v t + (1+v)^2 t^2 \tag{5.117}$$

and the expression under the square root factors as

$$u^2 - 4w^2 = (1 - (3v+2)t)(1 + (v+2)t)\,. \tag{5.118}$$

The first few terms of the square root are

$$\sqrt{u^2 - 4w^2} = 1 - vt - 2(1+v)^2 t^2 - 2v(1+v)^2 t^3 + \cdots \tag{5.119}$$

We can give a series representation of the square root via the series

$$\sqrt{1-z} = \sum_{n=0}^{\infty} \frac{(2n)!}{(1-2n)(n!)^2 4^n} z^n\,. \tag{5.120}$$

That is,

$$\begin{aligned}
&\sqrt{(1+(v+2)t)(1-(3v+2)t)} \\
&= \left(\sum_{n=0}^{\infty} \frac{(-1)^n(2n)!(v+2)^n t^n}{(1-2n)(n!)^2 4^n}\right)\left(\sum_{n=0}^{\infty} \frac{(2n)!(3v+2)^n t^n}{(1-2n)(n!)^2 4^n}\right) \\
&= \sum_{n=0}^{\infty} \frac{t^n}{4^n} \sum_{k=0}^{n} \frac{(-1)^k (2k)!(2n-2k)!}{(1-2k)(1-2n+2k)k!^2(n-k)!^2}(v+2)^k(3v+2)^{n-k}\,,
\end{aligned} \tag{5.121}$$

However, we are not sure if a closed form exists for these coefficients. Extracting x_2 coefficients from the overall expression is difficult because we would first need to expand it in series of t. Now, K_2 and K_2^* have $1+v$ in their denominators, so we can multiply the whole functional equation by $1+v$ to clear out these denominators. We have

$$(1+v)K_2 = c_2(1+v) - \overline{x}_2(1+t) = (c_2 - 1 - t)\overline{x}_2 + c_2 + c_2 x_2\,, \tag{5.122}$$

$$(1+v)K_2^* = c_2(1+v) - x_2(1+t) = c_2\overline{x}_2 + c_2 + (c_2 - 1 - t)x_2\,, \tag{5.123}$$

and

$$(1+v)\frac{\alpha(K_2^* - K_2)}{K_1} = \frac{\alpha(\overline{x}_2 - x_2)(1+t)}{K_1} = \alpha g(a_1, x_2, t)\,, \tag{5.124}$$

$$(1+v)\frac{\Delta_2(K_2^* K_{12})}{K_1} = c_2(\overline{x}_2 - x_2) f(a_1, x_2, t)\,, \tag{5.125}$$

where

$$f(a_1, x_2, t) = 1 - t\frac{c_1 - 1 + tv + \sqrt{(1+(v+2)t)(1-(3v+2)t)}}{c_1^2 - c_1 + c_1 tv + (1+v)^2 t^2}\,. \tag{5.126}$$

Then if we let $A = c_2 - (1+t)$, we can write the functional equation as

$$(c_2\overline{x}_2 + c_2 + Ax_2)Q_1^0(x_2) - (A\overline{x}_2 + c_2 + c_2 x_2)Q_1^0(\overline{x}_2) = c_2(\overline{x}_2 - x_2)f - \alpha g\,. \tag{5.127}$$

As before, we try to find a relation for the walks returning to the origin via coefficients extractions. We extract the x_2 coefficient from both sides to find that

$$c_2 Q_{1,2}^{0,2} + c_2 Q_{1,2}^{0,1} - (1+t)Q_{1,2}^{0,0} = -\alpha g_2^1 + c_2(f_2^2 - f_2^0)Q_{1,2}^{0,0}\,, \tag{5.128}$$

and then we set $a_2 = 1$ on both sides to find that

$$-(1+t)\breve{Q}_{1,2}^{0,0} = -\overline{a}_1 g_2^1\,. \tag{5.129}$$

When we finally substitute this quantity back into the equation, it simplifies to

$$b_2(Q^{0,2}_{1,2} + Q^{0,1}_{1,2}) = (b_2(f_2^2 - f_2^0) - a_2(1+t))Q^{0,0}_{1,2} - (1+t)\breve{Q}^{0,0}_{1,2}\,. \tag{5.130}$$

The question is can we find the quantity $f_2^2 - f_2^0$. It appears that

$$-(f_2^2 - f_2^0) = 1 + a_1 t + \frac{a_1^2}{b_1}(a_1^2 - a_1 - 4)t^3 + \frac{4a_1^2}{b_1^2}(a_1^3 - 3a_1^2 + 1)t^4 + \cdots\,, \tag{5.131}$$

which is not a series that would appear to have a closed form. Unfortunately, we have not been able to go further with this case. One could try making some assumptions for the form of equations like these for $Q^{m,n}_{1,2}$, $\breve{Q}^{m,n}_{1,2}$ and $\widehat{Q}^{m,n}_{1,2}$, but even then it is not clear if we can find a relation between $Q^{0,0}_{1,2}$, $\breve{Q}^{0,0}_{1,2}$, $\widehat{Q}^{0,0}_{1,2}$ and $\breve{\breve{Q}}^{0,0}_{1,2}$. We would have six equations like the one above plus six more, where $b_i = 0$ for the b_i's not present. Then one may use the four regular coefficient extractions from the master functional equation for $Q^{m,n}_{1,2}$ and $m, n \leqslant 1$ plus twelve more, where we set one or more of b_1 and b_2 to zero. We have a grand total of 28 equations in 36 unknowns, of which we need to eliminate 32. A quick attempt to do this via computer was not sufficient.

Chapter 6

Generalizations

In this final chapter we will analyze lattice walk models in more generality and we will see that a large class of models satisfy a functional equation of the type that we have seen. There is already some work done towards generalizations, but most of is on a small scale. For example, Definition 1.1 of [7, p. 2] has a functional equation for non-interacting lattice walks on the quarter plane with larger positive steps while the negative steps are limited to size at most 1. For quarter plane models with small steps where multiple copies of steps are allowed, see [11]. For some more general regions, check [21, 28, 29] for walks on a semi-infinite strip. In [27] the author considers lattice walks in an octant with interactions along the diagonal.

We will start by discussing marking operators in more generality. Then we will move onto finding functional equations for general models. Finally, we will look at some examples and ideas generated from them. Unfortunately, time did not suffice to explore this area to its full potential.

6.1 Marking Operators And Functional Equations

We start with the observation that operators that remove monomials are just marking operators with labels set to zero, and therefore, we will only consider marking operators for the rest of this work. A rather general operator is one that marks each point p in $\mathbb{Z}^d$ with a unique label a_p. This will serve as our most general definition. We will work with marking functions first and then express their action on monomials via operators.

Definition 6.1. *Given a set A of labels and constants from some space over indeterminates, let $a : \mathbb{Z}^d \to A$ be a marking function.*

Definition 6.2. *Given a set of labels A, let U_A be a set of marking functions on the space $\mathbb{Z}^d$. We define addition and multiplication term-wise on points of $\mathbb{Z}^d$, using the operations in the set A.*

Proposition 6.3. *The set of marking functions on $\mathbb{Z}^d$ together with addition and multiplication forms a commutative ring with identity. Furthermore, the ring is an integral domain. The set of invertible marking functions is comprised of functions with nonzero labels.*

Proof. If A is the whole space of labels and constants, the set U is the same as a Cartesian product of infinitely many copies of A, and it is not difficult to see that the properties of a commutative ring and an integral domain are satisfied. Here the additive identity is the marking function whose labels are all 0, and the multiplicative identity is one whose labels are all 1. Gotta think where these labels live in the end. □

Since we are working with marking operators on Laurent polynomials and series, we need a representation for marking operators that will allow us to compute things easily.

Definition 6.4. *Given a marking function a, let Γ_a be a marking operator defined by its action on monomials: $\Gamma_a x^p = a(p)x^p$ for all p in $\mathbb{Z}^d$. We extend this definition by linearity to formal Laurent series $u = \sum_p u_p x^p$, where $\Gamma_a u = \sum_p a(p)u_p x^p$, and we write $\Gamma_a = \sum_p a(p)[[x^p]]$.*

We need to check that this formal definition for marking operators behaves as expected and that it preserve the ring of functions U_A. Expanding the action $\Gamma_a u$ gives

$$\left(\sum_p a(p)[[x^p]]\right)\left(\sum_q u_q x^q\right) = \sum_{p,q} a(p)[[x^p]]u_q x^q = \sum_{p,q} a(p)u_q\delta_{p,q}x^q = \sum_p a(p)u_p x^p\,. \tag{6.1}$$

Furthermore addition and multiplication work as expected:

$$\Gamma_1 + \Gamma_2 = \sum_p a_1(p)[[x^p]] + \sum_p a_2(p)[[x^p]] = \sum_p (a_1(p) + a_2(p))[[x^p]]\,, \tag{6.2}$$

$$\begin{aligned}\Gamma_1\Gamma_2 &= \left(\sum_p a_1(p)[[x^p]]\right)\left(\sum_1 a_2(p)[[x^p]]\right)\\ &= \sum_{p,q} a_1(p)a_2(q)[[x^p]][[x^q]]\\ &= \sum_{p,q} a_1(p)a_2(q)\delta(p,q)[[x^p]]\\ &= \sum_p a_1(p)a_2(p)[[x^p]]\,.\end{aligned} \tag{6.3}$$

We just need the following notation to allow us to express certain operators easily.

Definition 6.5. *Let the term extraction operator be defined as*

$$[[x_I^P]] = \sum_{K \in \mathbb{Z}^{d-n}} [[x_{I,J}^{P,K}]] \,, \tag{6.4}$$

where $I = \varnothing$ gives the identity operator $1 = \sum_p [[x^p]]$

Proposition 6.6. *The set of marking operators Γ_a for a from U_A with addition and multiplication, as defined in the preceding discussion, forms a commutative ring with identity. Furthermore, the ring is an integral domain.*

Lemma 6.7. *The marking operator Γ_a, as defined by a sum, is well-defined and has the secondary form $\Gamma = 1 + \sum_p (a(p) - 1)[[x^p]]$, where the sum runs over all points in $\mathbb{Z}^d$ with $a(p) \neq 1$.*

Proof. Assume that another operator Γ' behaves in the same way as in the definition. Then

$$\begin{aligned} \Gamma u - \Gamma' u &= \left(\sum_p a(p)[[x^p]] \right) \left(\sum_q u_q x^q \right) - \sum_p a(p) u_p x^p \\ &= \sum_p a(p) u_p x^p - \sum_p a(p) u_p x^p = 0 \,, \end{aligned} \tag{6.5}$$

and so the operator is well-defined. The second form follows by writing $\Gamma = 1 + (\Gamma - 1)$ and expanding the latter quantity. □

The following lemma is self-evident.

Lemma 6.8. *For each point p, the monomial x^p is an eigenfunction of Γ_a with eigenvalue $a(p)$.*

Corollary 6.9. *The eigenspaces of formal Laurent series for each eigenvalue λ of Γ_a are generated by sums of monomials x^p with $a(p) = \lambda$.*

Proof. Let $\Gamma u = \lambda u$ for some formal Laurent series. Then the only surviving terms in $\Gamma u - \lambda u$ are those with $a(p) \neq \lambda$. However, since the quantity is zero, they are also forced to add up to zero. Therefore, the series u is comprised of terms with monomials x^p for $a(p) = \lambda$ in the first place. □

Corollary 6.10. *The space of Laurent series annihilated by Γ_a is generated by series of terms x^p with zero labels.*

For all the problems so far, we have had marking operators comprised of finitely many term extractions and the identity 1, where each term extraction has less variables than the dimension d. As the following example shows, sometimes a marking operator comprised of infinitely many term extractions may be rewritten in terms of finitely many term extractions with less variables.

Example 6.11. For example, consider the operator $\Gamma = \sum_p a[[x^p]]$ where all points receive the same label. Then $\Gamma = a$.

Sometimes, depending on the model, we may even have a smaller operator that does the same work as the full marking operator because of the geometry of the regions involved.

Example 6.12. Consider a one-dimensional classic model with no interactions. Because we do not want lattice walks to step on negative coordinates, the full marking operator is given by

$$\Gamma = 1 - \sum_{n=1}^{\infty} [[\overline{x}^n]] \,. \tag{6.6}$$

However, since the lattice walk may never reach negative coordinates beyond negative one, we have the equivalent form $\Gamma' = 1 - [[\overline{x}]]$. Although the operators are different in form, they are essentially the same with regards to the problem at hand. We will say the two operators are model-equivalent.

To make the equivalence work, we need to consider the whole model. With that in mind, we may think of the lattice walk models as living completely within the sumset $\mathcal{D} + \mathcal{S}$. This is how far walks from the model can possibly reach from within $\mathcal{D}$ by taking a single step. We will call this sumset the range of the model. In the enumeration of lattice walks, we must eliminate all walks that take a step outside of $\mathcal{D}$, but they remain in $\mathcal{D} + \mathcal{S}$, so it is enough to remove walks in the complement of $\mathcal{D}$ inside the sumset. Note that the walks do not necessarily reach all points in $\mathcal{D} + \mathcal{S}$ as one can see in the case of reducible models.

Definition 6.13. *Given sets A and B, we let their sumset be*

$$A + B = \{a + b \,|\, a \in A,\, b \in B\} \,. \tag{6.7}$$

Definition 6.14. *Given a lattice walk model $\mathcal{M}$ with domain $\mathcal{D}$ and step set $\mathcal{S}$, define the range of model $\mathcal{M}$ to be the sumset $\mathcal{D} + \mathcal{S}$.*

Definition 6.15. *Let $\mathcal{M}$ be a lattice walk model in $\mathbb{Z}^d$ and let Γ and Γ' be two marking operators. Then $\Gamma \equiv_{\mathcal{M}} \Gamma'$ if and only if Γ and Γ' act identically on monomials from the range of $\mathcal{M}$. That is, the operators are model-equivalent if and only if $\Gamma x^p = \Gamma' x^p$ for all $p \in \mathcal{D} + \mathcal{S}$.*

In the preceding example, the operators are equivalent because they act the same way on monomials from the range of the model, that is, monomials x^k with $k \geqslant -1$. Similarly, in the two-dimensional problem we have an equivalence between the operator E and the operator $E' = (1 - [[\overline{x}_1]])(1 - [[\overline{x}_2]])$.

Definition 6.16. *A marking operator Γ is finite with respect to a given model if it is equivalent to an operator Γ' comprised of finitely many terms. That is,*

$$\Gamma \equiv_{\mathcal{M}} \Gamma' = \sum_{\substack{I \subseteq [d] \\ J \subseteq \mathbb{Z}^d}} a_{I,J}[[x_I^J]], \tag{6.8}$$

where each set J is finite and $a_{I,J}$ is a label. Also, we say Γ is infinite if it is not finite.

Lemma 6.17. *A finite product of finite marking operators is finite.*

Definition 6.18. *Given a d-dimensional model, let L be an operator that marks locations in $\mathcal{D}_0$ and let E be the operator that removes walks outside of $\mathcal{D}$. Then the marking operator corresponding to the model is $\Gamma_a = LE$ with marking function a which sends points p in $\mathcal{D}_0$ to their respective labels, and points outside of $\mathcal{D}$ to zero.*

When we work within the framework set by a specific model, we will not distinguish between Γ and Γ'. A natural question is to consider which models lead to finite operators. This depends on the region $\mathcal{D}$ and the nature of the marking of region $\mathcal{D}_0$. We have some cases where it is easy to say.

Lemma 6.19. *Let $\mathcal{D}$ be a finite box and let $\mathcal{S}$ be finite. Then the marking operator $\Gamma = LE$ is finite.*

Proof. The operator L is finite since it marks only finitely many points inside the box. We will find E next. Let the box have x_k coordinates in $[a_k, b_k]$. Suppose that the x_k coordinates of the steps are bounded in size by n_k. If $n_k = 0$, the walks do not leave the $x_k = a_k$ hyperplane and there is nothing to remove. When $n_k > 0$, we need to remove all monomials $x_1^{P_1} \cdots x_d^{P_d}$ with $P_k \in [a_k - n_k, a_k - 1] \cup [b_k + 1, b_k + n_k]$. Then $E = E_1, \ldots, E_d$, where

$$E_k = 1 - \sum_{j=1}^{n_k} [[x_k^{a_k - j}]] + [[x_k^{b_k + j}]], \tag{6.9}$$

and $E_k = 1$ when $n = 0$. Each of these is finite and so is E. □

Note that the operators E_k remove monomials outside of the range of the model too, but that allows us to write the operator down in a neater way.

Theorem 6.20. *Consider a model where $\mathcal{D}$ and $\mathcal{S}$ are finite. Then the operator $\Gamma = LE$ is finite.*

Proof. The region $\mathcal{D}$ lies within a bounding box B, so we may write $E = E'E''$, where $E' = E_1 \cdots E_d$ from the lemma and E'' is the operator removing all monomials outside of $\mathcal{D}$ but inside the interior of B. E'' is finite since there are only finitely many points in that region. Since we are marking only finitely many points, L is finite too, and finally, Γ is finite. □

Also, we may handle partially-infinite regions, corresponding to boxes extended to infinity on some sides, if we are marking only finitely many points. Here one or more of a_k and b_k is infinite and we actually lose parts of E_k since no terms need to be removed at infinity.

Definition 6.21. *A lattice walk model with finitely many steps is finite if its marking operator $\gamma = LE$ is finite, and its initial condition q has finitely many terms. Otherwise, the model is infinite.*

Example 6.22. All the models so far have been finite, but not all models are finite. Consider a one-dimensional model on the whole line, where for each integer i, we mark point x^i with parameter a_i using operator L. Then all equivalent marking operators L' would have to have a term like $(a_i - 1)[[x^i]]$; otherwise, it would not act the same way on monomials in the model's range $\mathbb{Z}$. Therefore, L is not equivalent to a finite operator.

Finite models are nicer because they have functional equations with finitely many terms and there is a chance that they will be amenable to the sort of analysis we have performed for quarter plane walk models in the past.

Theorem 6.23. *The generating function Q for a lattice walk model satisfies the equations*

$$(1 - tLES)Q = q\,, \tag{6.10}$$

$$(L^{-1} - tES)Q = L^{-1}(q) = \alpha\,. \tag{6.11}$$

Proof. Because of how we have set up the notation so far, the proof is analogous to the one in Chapter 2 and Chapter 3. The second functional equation is obtained by applying L^{-1} to the first one, so it only remains to show that the first functional equation holds. When we apply LE to tSQ we remove all monomials for walks ending outside of $\mathcal{D}$ and we mark all monomials for walks ending in $\mathcal{D}_0$. We have almost obtained Q again, but the first term $q_0 = q$ is missing, so we add it in. That is $LE(tSQ) + q = Q - q + q = Q$ and we are done. □

Corollary 6.24. *When the model is finite, we may perform the coefficient extractions in the functional equations to obtain explicit equations relating Q to various Q_I^J terms.*

This follows because there are only finitely many extractions in the marking operators. It is not clear if may have explicit functional equations for models with infinite operators because there would be infinitely-many Q_I^J terms in the equation. We would need more theory in order to do that.

6.1.1 Several Examples of Generalized Models

In the first two examples, we will see some more general analysis on models. After that we will consider some more concrete situations.

Example 6.25. Typically, we have an operator $\Gamma = LE$ where E removes unwanted terms. The problem is that E is not invertible. We can recast the analysis in a way that circumvents that but at some cost. Instead of using an operator E to remove walks, we may just just a marking operator that labels walks for later deletion, for example, with a variable z. We have a new lattice walk model $\widetilde{\mathcal{M}}$ with $\widetilde{\mathcal{D}} = \mathbb{Z}^d$ and $\widetilde{\mathcal{D}}_0 = \mathcal{D}_0 \cup \mathcal{D}^c$, where we mark monomials in $\mathcal{D}^c$ with z and those in $\mathcal{D}_0$ in the same way as in model $\mathcal{M}$. Here $\widetilde{E}$ is the identity operator since we do not remove any walks. We have the functional equations

$$(1 - t\widetilde{L}S)\widetilde{Q} = q\,, \tag{6.12}$$

$$(\widetilde{L}^{-1} - tS)\widetilde{Q} = \widetilde{\alpha}\,. \tag{6.13}$$

The downside is that we have more extractions than if we split the operator as $\Gamma = LE$ and only applied L^{-1}. If we obtained a solution for $\widetilde{Q}$, then we could find the original Q by taking $Q = [z^0]\widetilde{Q}$.

Example 6.26. We can also keep track of the walks' histories of where they have been. To do this, we let Γ give a generic label to each point in space, that is, $\Gamma x^p = a_p$. In this setup, a growing lattice walk picks up a label at each point that it steps on. As in Example 6.22, the operator is not finite, so we would not have an explicit functional equation with finitely many terms. Nevertheless, this allows us to frame self-avoiding within our theory.

Suppose that Q is the generating function for walks with histories and let W be the generating function for self-avoiding walks on the same space. Then we can obtain W from Q by setting all higher powers of a_p to zero. This is equivalent to picking out the a_p^0 and a_p coefficients from Q via some operator A. Since there are only finitely many walks of a given length and they are bounded, for each step size, A reduces to a finite operator A_k, that is, $W = AQ = \sum_{k=0}^{\infty} A_k(q_k)t^k$.

The following examples will look at particular models with small steps in the plane.

Example 6.27. Consider an infinite strip comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant j \leqslant n$, where the model interacts with the lines $x_2 = 0$ and $x_2 = n$. The operators are

$$E = E_{-1}E_{n+1} = (1 - [[\overline{x}_2]])(1 - [[x_2^{n+1}]]) = 1 - [[\overline{x}_2]] - [[x_2^{n+1}]]\,, \tag{6.14}$$

$$L = L_0L_n = (1 + b_0[[x_2^0]])(1 + b_n[[x_2^n]]) = 1 + b_0[[x_2^0]] + b_n[[x_2^n]]\,, \tag{6.15}$$

and the functional equation for Q is given by

$$KQ - (c_0 - t\overline{x}_2S_2^{-1})Q_2^0 - x_2^n(c_n - tx_2S_2^1)Q_2^n = \overline{a}_0\,. \tag{6.16}$$

If we extract the $[x_2^0]$ term, we find that

$$(1 - a_0 t S_2^0) Q_2^0 - t S_2^{-1} Q_2^1 = 1 \tag{6.17}$$

and if we extract the $[x_2^n]$ term, we find that

$$(1 - a_n t S_2^0) Q_2^n = t S_2^1 Q_2^{n-1} \,. \tag{6.18}$$

Similarly, we can extract the $[x_2^k]$ terms for $1 \leqslant k \leqslant n-1$. We will have $n+1$ equations where we need to solve for the $n+1$ unknown functions $Q_2^0, \ldots, Q_2^n$. Of course, we only need Q_2^0 and Q_2^n to obtain a solution for Q. In the simplest case, $n = 1$ and the term extraction equations become

$$(1 - a_0 t S_2^0) Q_2^0 = 1 + t S_2^{-1} Q_2^1 \,, \tag{6.19}$$

$$(1 - a_1 t S_2^0) Q_2^1 = t S_2^1 Q_2^0 \,. \tag{6.20}$$

The solutions are

$$Q_2^0 = \frac{1 - a_1 t S_2^0}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2} \,, \tag{6.21}$$

$$Q_2^1 = \frac{S_2^1 t}{(1 - a_0 t S_2^0)(1 - a_1 t S_2^0) - S_2^1 S_2^{-1} t^2} \,, \tag{6.22}$$

and we can find Q explicitly from

$$Q = \frac{1}{K}(\overline{a}_0 + (c_0 - t\overline{x}_2 S_2^{-1}) Q_2^0 + x_2 (c_1 - t x_2 S_2^1) Q_2^1) \,. \tag{6.23}$$

Example 6.28. In the plane, consider a finite rectangle comprised of all the points with coordinates (i, j) in $\mathbb{Z}^2$ with $0 \leqslant i \leqslant m$ and $0 \leqslant j \leqslant n$, for some positive integers m and n. There are also interactions like in the previous example. We will use the operators

$$E_{i,j} = 1 - [[x_i^j]] \quad \text{and} \quad L_{i,j} = 1 + b_{i,j} [[x_i^j]] \,. \tag{6.24}$$

to find the functional equation. For the model itself, we have

$$E = E_{1,\text{-}1} E_{1,m+1} E_{2,\text{-}1} E_{2,n} \,, \tag{6.25}$$

$$L = L_{1,0} L_{1,m} L_{2,0} L_{2,n} \,. \tag{6.26}$$

Once we obtain a functional equation, we will find that it has terms of the form Q_i^j and $Q_{1,2}^{i,j}$. If we perform the extractions $[x_i^j]$ only we will have even more Q terms. However, if we perform the

extractions $[x_i^j]$ for all points (i, j) in the rectangle, we will obtain $(m+1)(n+1)$ equations in that many $Q_{1,2}^{i,j}$ terms. Since there is a unique solution Q, we can solve this system and write Q directly. The difficulty lies is in solving this system symbolically.

This example paves the way towards the following result for finite models on finite regions.

Theorem 6.29. *The generating function for a finite lattice walk model set on a finite region is a rational function in its variables*

Proof. Let n be the number of points in $\mathcal{D}$. Then for each point P, we may extract the $x_{[d]}^P$ term from the functional equation $(1 - t\Gamma S)Q = q$ and we will only have Q terms of the form $Q_{[d]}^I$ for points I in $\mathcal{D}$. We have a system of n equations in n unknowns, and since the function Q is unique, we may solve this system uniquely. Each of the coefficients is a rational function of the variables $x_1, \ldots, x_d$, the labels a_P, and t, and therefore, the solution is rational too. □

Of course, once the region grows to even a moderate size, it becomes prohibitively complicated to solve this system. For a box of side m, the number of equations is m^d, making this a rather difficult way of solving the problem. If the size of the region is large and the number of steps in S is small, then each $Q_{1,2}^{i,j}$ depends on a few neighbours only, so the resulting linear system will be rather sparse. Still, the solution fills a whole page for a three-by-three square, so this result is mostly of theoretical value.

6.2 Further Properties of Marking Operators

Since the function Q is given by Neumann series **[Bonus: Write more about this in this chapter]**, which is connected to integral equations, it may be possible to find an analogue to Fredholm theory. Also, it may be possible to find a sort of Sturm-Liouville theory for lattice walk enumeration problems. In the next few pages we will look at the first few steps of such a theory, although there was insufficient time to analyze this problem in more detail. For that we would need a suitable inner product that would allow us to see if certain operators are self-adjoint. Finally, we would find eigenfunction expansions for the functions involved.

The operators we have seen so far are marking operators, like L and E, in combination of multiplication by another function, like ΓS, or a combination of operators, like $L^{-1} - tES$. We will look at their eigenvalues, eigenvectors, kernels and invertibility.

6.2.1 Marking Operators With Multiplication

Definition 6.30. *We call an operator inhomogeneous if it consists of a homogeneous operator* Γ *and a Laurent polynomial* S*, and acts in one of the following ways*

$$\Gamma S(u) = \Gamma(Su) \quad \textit{or} \quad S\,\Gamma u = S \cdot \Gamma u\,. \tag{6.27}$$

The two definitions are equivalent when it comes to finding the eigenpairs.

Proposition 6.31. *The pair* (λ, u) *is an eigenpair for* ΓS *if and only if* (λ, Su) *is an eigenpair for* $S\,\Gamma$*. In particular, if* $\lambda = 0$*, then* $\ker(\Gamma S) = \overline{S}\ker(\Gamma)$*.*

Proof. Note that $\Gamma S u = \lambda u$ is equivalent to $S\,\Gamma(Su) = \lambda(Su)$. Therefore, the operators have the same eigenvalues and their eigenfunctions are u and Su, respectively. Next, we pick $\lambda = 0$ to find the kernels of the operators. We find that $u \in \ker(\Gamma S)$ if and only if $Su \in \ker(S\,\Gamma) = \ker(\Gamma)$. □

Since this characterizes the kernel of the operator, we will assume that $\lambda \neq 0$ from here on. We will now work towards finding some eigenpairs. Let $\Gamma = LE$ as before. Recall that $E^2 = E$, since once the terms are removed, they cannot be removed again. We can apply E to both sides of $LESu = \lambda u$ to find that

$$E(\lambda u) = \lambda E u = ELESu = LE^2 Su = LESu = \lambda u \tag{6.28}$$

and therefore, $Eu = u$.

Proposition 6.32. *If* u *is an eigenfunction of the operator* LES*, then* $Eu = u$ *and* $u \in \ker E'$*, where* $E' = E - 1$ *is the pure coefficient extraction part of* E*.*

Suppose that $\Gamma = LE$. In the interest of performing fewer extractions, we rewrite the eigenproblem $\Gamma S u = \lambda u$ as $ESu = \lambda M u$, where $M = L^{-1}$. Since each operator is the identity plus a sum of extractions, we rewrite the operators as $E = E - 1 + 1 = 1 + E'$ and $M = 1 + M'$, where E' and M' are purely extractions. Using these relations, we can rewrite the function u as

$$u = \frac{(\lambda M' - E'S)u}{S - \lambda}\,, \tag{6.29}$$

where if the operators are finite, the extraction in the numerator leads to a Laurent polynomial in the variables $x_1, \ldots, x_d$. Because we used M' and E' instead of the equivalent for LE, we have fewer coefficients in the numerator. The resulting eigenfunction u depends on finitely many coefficients. We may expand u in Laurent series around the origin, set the coefficients to zero and

then solve the resulting system of equations to find the terms in the numerator. Curiously, if we apply $D = \lambda M - ES$ to u we should find that $Du = 0$, but here u has a new form and we will obtain some condition on the terms in the numerator.

Example 6.33. Consider one of the one-dimensional classic model with interactions, where $S = \overline{x} + x$ and $M' = -c[x^0]$. First we will use the equivalent operator $E' = -\sum_{k=1}^{\infty}[[\overline{x}^k]]$. Since u has no terms u_k for $k \leqslant$ -1, we see that

$$-E'Su = \sum_{k=1}^{\infty} \overline{x}^k([\overline{x}^{k+1}] + [\overline{x}^{k-1}])u = \sum_{k=1}^{\infty} \overline{x}^k(u_{-k-1} + u_{-k+1}) = \overline{x}u_0\,. \tag{6.30}$$

Then

$$u = \frac{-\lambda c u_0 + \overline{x}u_0}{\overline{x} + x - \lambda} = \frac{(1 - c\lambda x)u_0}{1 - \lambda x + x^2} = u_0 + \overline{a}\lambda u_0 x + \cdots \tag{6.31}$$

where u_0 is a free variable, and we have an eigenfunction for every λ. Next, we will use the equivalent operator $E = -[[\overline{x}]]$. We have no reason to believe that the operator will have the same eigenfunctions, but it does. Here we have

$$u = \frac{(\lambda M' - E'S)u}{S_\lambda} = \frac{-c\lambda u_0 + (u_{-2} + u_0)\overline{x}}{\overline{x} + x - \lambda} = \frac{u_{-2} + (1 - c\lambda x)u_0}{1 - \lambda x + x^2}\,, \tag{6.32}$$

We can apply the operator D on u to find that $-b\lambda u_{-2} = 0$, so $u_{-2} = 0$ and the eigenfunctions are the same.

It turns out that this is not a coincidence and we have the following striking result.

Proposition 6.34. *The set if eigenpairs of the full marking operator associated with a lattice walk model is a subset of the set of eigenpairs for any equivalent but reduced operator. Here reduced means a change to the marking portion and a reduction of terms in the extraction portion.*

Proof. Let $\Gamma_1 = L_1E_1$ be the full operator and $\Gamma_2 = L_2E_2$ be the reduced one. Here L_1 and L_2 agree on $\mathcal{D}$, and E_1 and E_2 agree on the range $\mathcal{D} + \mathcal{S}$. Suppose that $\Gamma_1 Su = \lambda u$ for some λ and u. Since u contains terms from $\mathcal{D}$ only (by Proposition 6.32), then Su contains terms from the range of the model. Therefore, $\Gamma_1 Su = L_1E_1Su = L_2E_2Su$ due to the agreement of operators. Therefore, $\Gamma_2 Su = \lambda u$ too. □

The proof here also shows us that Γ_2 may have other eigenfunctions v, since here v may contain terms from outside of the range of the model. However, if Γ_2 is finite, we see that $(\lambda M' - E'S)u$ has finitely many terms. Thus each eigenvalue λ may lead to at most a finite-dimensional eigenspace.

6.2.2 Lattice Walks Operator

In this section we explore the operator $\mathcal{L} = 1 - t\Gamma S$ for a step Laurent polynomial S and some marking operator Γ. Let (u, λ) be an eigenpair for $\mathcal{L}$. Then we calculate

$$\begin{aligned}(\mathcal{L} - \lambda I)u &= u - \lambda u - t\Gamma S u \\ t\Gamma S u &= (1-\lambda)u\,.\end{aligned} \tag{6.33}$$

If $\lambda = 1$, then u is in $\ker(t\Gamma S) = \overline{S}\ker\Gamma$ and we already know how to find these. If $\lambda \neq 1$, we may find the eigenpairs by the method in the previous section. Let $\mu = 1 - \lambda$. Then

$$u = \frac{((1-\lambda)M' - tE'S)u}{\lambda - K}\,, \tag{6.34}$$

where $K = 1 - tS$.

Chapter 7

Epilogue

In this thesis we looked at the problem of generalizing methods for handling lattice walk enumeration in many dimensions and with more general constraints, as well as models with interactions. In the earlier chapters we used refined notation and a new perspective of applying operators to functions and functional equations. The result was that we were able to derive the functional equations for d-dimensional classic models with ease. This allowed us to generalize the methods to derive a functional equation for a much more general class of models. This is one more step towards finding out if we may automatize the solutions and classification of more complex lattice walk models. In that vein, we were able to extend the algebraic and obstinate kernel methods to lattice walk models with interactions, although there is still a fair amount of work that needs to be done. First, the algebraic kernel method does not work in a lot of cases of finite groups when the model has interactions, and in those cases that it does work, it does not manage to eliminate the $Q^{0,0}_{1,2}$ term. The resulting formulae still need to be worked through. At best, we hope that in those cases, if the function $Q^{0,0}_{1,2}$ is D-finite, then the full generating function Q is D-finite too. In the extension of the obstinate kernel method, we were able to find relations between $Q^{0,0}_{1,2}$ and its various partially-marked versions for two models - the Cartesian and Diagonal models. However, we were not able to crack the Full model. It also remains to be seen if the cases with less transformations preserving the admissibility of the functions Q^0_i can also be tackled. The problem of solving the functional equation for more generalized models is wide open.

7.1 Future Research

The future research can be broken down into several over-arching ideas.

Algebraic kernel method

- In the cases in this thesis, determine we can deduce information about the D-finiteness of Q with respect to the D-finiteness of $Q_{1,2}^{0,0}$.
- Determine if it can be extended to higher-dimensional classic models with interactions.
- Find out if the method can be extended to models with larger steps.
- Consider other operators that may annihilate terms of the functional equation and find out if this can be used to find out more information about the generating functions.

Obstinate kernel method

- Explain combinatorially the identities between the marked and partially-marked version of $Q_{1,2}^{0,0}$. A good place to start would be the one-dimensional models.
- In the diagonal model, find a solution for Q in terms of partially-marked versions of Q and $Q_{1,2}^{0,0}$.
- Find out if there is a similar relationship for $Q_{1,2}^{0,0}$ in the full model, and in that direction, find out if we can deal with square roots in the extractions of the reduced functional equations.
- Solve some of the enumeration problems with other groups, like those of order six.

Kernel methods in general

- Determine if the algebraic and obstinate kernel methods can be combined into one holistic approach. This may involve classifying methods on the number of substitutions made.
- Find out the methods can be extended for more generalized models, as in the previous chapter.

Connection with other theory

- Using the Cauchy integral formula, we may rewrite coefficient extractions as complex integrals. This suggests that some of the functional equations may be turned into multidimensional Fredholm integral equations. Perhaps methods from that area can be adapted to the enumeration of lattice walk models. Another clue is the Neumann series solution, which is employed in both theories.

- Explore a kind of a Sturm-Liouville theory approach to solving lattice walk enumeration problems. Since we are working with operators and functional equations, it is perhaps possible to find some suitable inner products, where the operators will be self-adjoint. We hope that solutions of the functional equations may be written in terms of eigenfunction expansions of the operators involved.
- Can the functional equations with operator form be used to find asymptotics for the generating functions involved?

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